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AIUB, Data Communication Notes

DATA COMMUNICATION

Table of Contents

ANALOG TRANSMISSION

Digital to Analog Conversion

- **Digital-to-Analog Conversion (DAC)** changes an analog signal's characteristics using digital data.
- It modifies parameters like amplitude, frequency, or phase.
- The result is an **analog signal** that carries the original digital information.

⇨ Types of digital to analog conversion:

1. **Amplitude Shift Keying (ASK)**
    - Varies the **amplitude** of the carrier signal based on digital data.
    - Example: '1' = high amplitude, '0' = low amplitude.

2. **Frequency Shift Keying (FSK)**
    - Varies the **frequency** of the carrier signal.
    - Example: '1' = high frequency, '0' = low frequency.

3. **Phase Shift Keying (PSK)**
    - Varies the **phase** of the carrier signal.
    - Example: '1' = 0° phase, '0' = 180° phase shift.

4. **Quadrature Amplitude Modulation (QAM)**
    - Combines **ASK & PSK**.
    - Sends more bits per symbol by varying both amplitude and phase.

Basic of Conversion

- **Bit Rate:** Number of bits transmitted per second (**bps**).
- **Baud Rate:** Number of signal elements (modulation changes) per second.
- **Carrier Signal:** A high-frequency base signal used in analog transmission to carry digital data.

Analog transmission

- **Formula:**

$$\textbf{Signal rate } S = N \times \frac{1}{r} \ \textbf{ baud}$$

Where:

- N = Bit rate (bps)
- **r** = number of bits per signal element

$$r = log_2 L$$

- L = number of signal elements

In analog transmission of digital data, **baud rate ≤ bit rate**.

Problem1: An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate.

➔ **Is given,**

S = 1000 bound

r = 4

We Know,

$$S = N \times \frac{1}{r}$$

$$\therefore N = S \times r = 1000 \times 4 = 4\text{kbps}$$

Problem2: An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

➔ **Is given,**

S = 1000 bound

N = 8000 bps

We Know,

$$S = N \times \frac{1}{r}$$

$$\therefore r = N \times \frac{1}{S} = 8000 \times \frac{1}{1000} = 8 \text{ bits/bound}$$

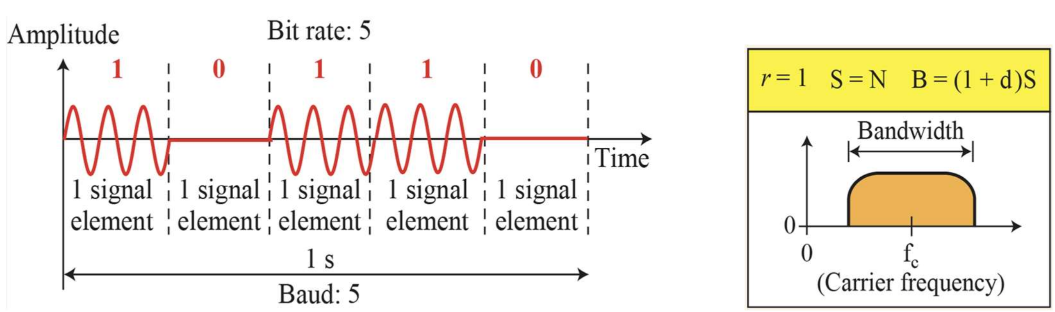
$$\therefore L = 2^r = 2^8 = 256$$

Binary Amplitude Shift Keying

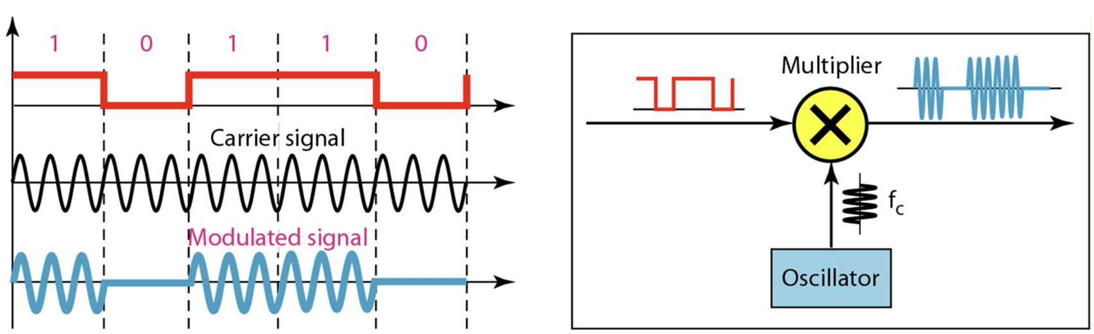
- Varies **amplitude** of carrier signal.
- **Frequency** and **phase** stay constant.
- **Binary ASK (On-Off Keying or OOK)**:
    - '1' = signal with amplitude
    - '0' = no signal (zero amplitude)

'1' = signal with amplitude

'0' = no signal (zero amplitude)



Implementation of binary BASK



Formulas

Bandwidth

Bandwidth:	Max Bandwidth: (d=1)
$$\mathbf{B = (1 + d) \times S}$$ Where: • **B** = Bandwidth required (in Hz) • **S** = Signal (baud) rate • **d** = **Modulation factor** or **roll-off factor** (0~1)	$$\mathbf{B_{max} = 2S}$$ **Max Bandwidth: (d=0)** $$\mathbf{B_{min} = S}$$

Carrier Frequency

Carrier Frequency:

$$\boldsymbol{f_c = \frac{f_{max} + f_{min}}{2}}$$

Problem3: We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using BASK with d = 1?

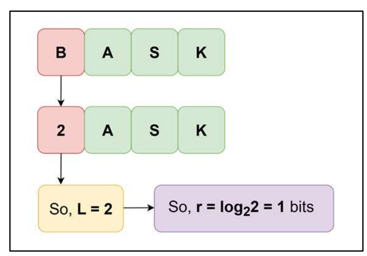
➔ **Is given,**

B = 100k Hz

$f_{max} = 300\ kHz$

$f_{min} = 200\ kHz$

$d = 1 \qquad r = 1$



We Know,

$$B = (1 + d) \times S$$

[since $S = N \times \frac{1}{r}$]

$$B = (1 + d) \times N \times \frac{1}{r}$$

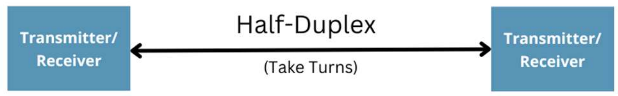
$$\therefore \text{N} = \frac{B}{(1+d)} = \frac{100K}{1+1} = 50 \text{ Kbps}$$

carrier frequencies:

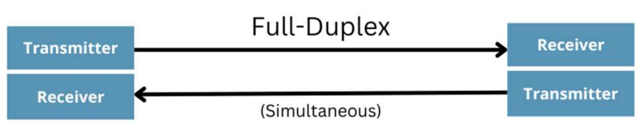
$$f_c = \frac{f_{max} + f_{min}}{2} = \frac{300k + 200k}{2} = 250k\ Hz$$

Half Duplex & Full Duplex

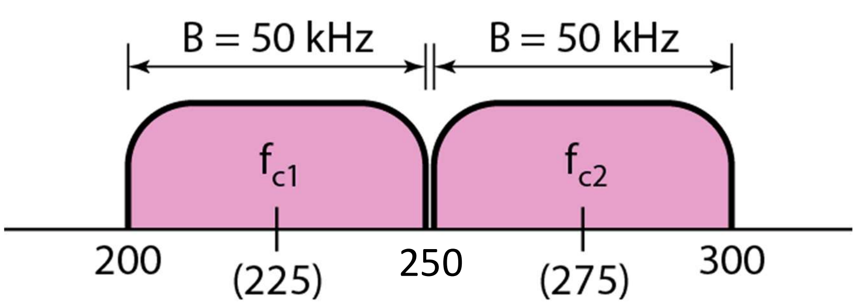
- **Half Duplex**: Data transmission occurs in both directions, but only one direction at a time. Example: walkie-talkies.



- **Full Duplex**: Data transmission occurs simultaneously in both directions. Example: telephone calls.



  - **Bandwidth Split**: Total bandwidth is divided into two using separate carrier frequencies.
  - **Example**: 100 kHz total → 50 kHz per direction.
  - **Data Rate**: 25 kbps in each direction.



Problem4: We have an available bandwidth of 100 kHz, which spans from 200 to 300 kHz. If we implement full duplex communication using Amplitude Shift Keying (ASK) with a distance factor d =1 and a rate factor r =1, what are the **carrier frequencies** and the **bit rates** for both the uplink and downlink channels?

➔ **Is given,**

$B_{total} = 100\text{k Hz}$

$d = 1 \qquad r = 1$

➔ Since it's **full duplex**, divide total bandwidth equally:

$B_{up} = 50\text{ kHz}$

$B_{down} = 50\text{ kHz}$

For uplink:

$f_{min} = 200\ kHz$

$f_{max} = 250\ kHz$

For downlink:

$f_{min} = 250\ kHz$

$f_{max} = 300\ kHz$

We Know,

Bit rates:

$$B_{up} = (1+d) \times S \qquad [\text{since } S = N \times \frac{1}{r}]$$

$$B_{up} = (1+d) \times N \times \frac{1}{r}$$

$$\therefore \text{N} = \frac{B_{up}}{(1+d)} = \frac{50\text{K}}{1+1} = 50 \text{ Kbps} = 25 \text{ kbps}$$

$$B_{down} = (1+d) \times S \qquad [\text{since } S = N \times \frac{1}{r}]$$

$$B_{down} = (1+d) \times N \times \frac{1}{r}$$

$$\therefore \text{N} = \frac{B_{down}}{(1+d)} = \frac{50\text{K}}{1+1} = 50 \text{ Kbps} = 25 \text{ kbps}$$

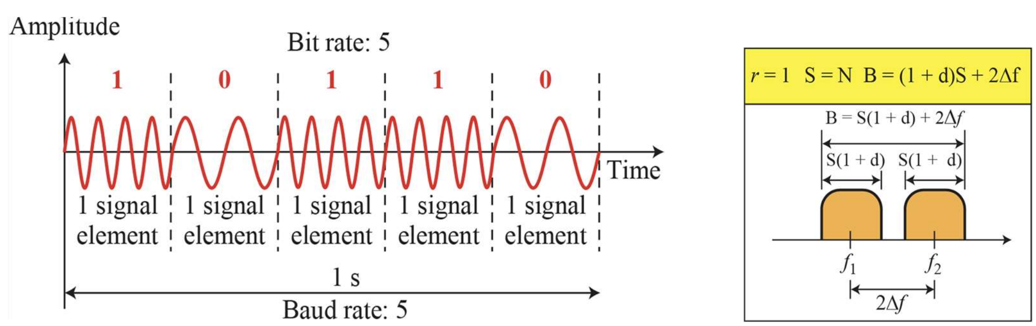
carrier frequencies:

$$f_{c_{up}} = \frac{f_{max} + f_{min}}{2} = \frac{250k + 200k}{2} = 225k\ Hz$$

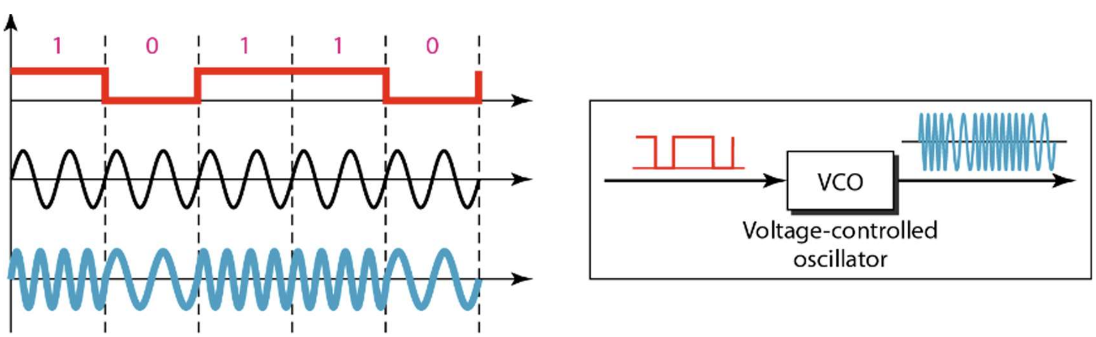
$$f_{c_{down}} = \frac{f_{max} + f_{min}}{2} = \frac{300k + 250k}{2} = 275k\ Hz$$

Binary Frequency Shift Keying

- In **FSK (Frequency Shift Keying)**, data is represented by **changing the carrier frequency**.
- Each signal element uses a **constant frequency**.
- Frequency **changes only if the data bit changes**.
- **Amplitude and phase** remain **constant**.



Implementation of binary BFSK



Formulas

Bandwidth

Bandwidth: $$\boxed{B = (1 + d) \times S + 2\Delta f}$$ Where: • **B** = Bandwidth required (in Hz) • **S** = Signal (baud) rate • **d** = **Modulation factor** or **roll-off factor** (0~1) • **2Δf = difference** between two **Carrier Frequency**	**Max Bandwidth: (d=1)** $$B_{max} = 2S + 2\Delta f$$ **Max Bandwidth: (d=0)** $$B_{min} = S + 2\Delta f$$ $$2\Delta f = f_{c1} - f_{c2}$$

Bandwidth (when roll-off factor d is not given):

$$\boxed{B = L \times S}$$

Where:

- **L** = Number of signal levels
- **S** = Signal rate (baud)

Carrier Frequency

Carrier Frequency:

$$\boxed{f_c = \frac{f_{max} + f_{min}}{2}}$$

Problem5: We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with d = 1?

➔ **Is given,**

B = 100k Hz

$f_{max} = 300\ kHz$

$f_{min} = 200\ kHz$

$d = 1 \qquad r = 1$

➔ Since it's **FSK**, divide total bandwidth equally:

B_{up} = 50 kHz

B_{down} = 50 kHz

For uplink:

$f_{min} = 200\ kHz$

$f_{max} = 250\ kHz$

For downlink:

$f_{min} = 250\ kHz$

$f_{max} = 300\ kHz$

We Know,

carrier frequencies:

$$f_{c_{up}} = \boldsymbol{f_{c1}} = \frac{f_{max} + f_{min}}{2} = \frac{250k + 200k}{2} = 225k\ Hz$$

$$f_{c_{down}} = \boldsymbol{f_{c2}} = \frac{f_{max} + f_{min}}{2} = \frac{300k + 250k}{2} = 275k\ Hz$$

$$\therefore\ \mathbf{2\Delta f} = \boldsymbol{f_{c2} - f_{c1} = 275K - 225K = 50K\ Hz}$$

Bandwidth:

$B = (1 + d) \times S + 2\Delta f$ [since $S = N \times \frac{1}{r}$]

$B = (1 + d) \times N \times \frac{1}{r} + 2\Delta f$

$\therefore$ N $= \frac{B - 2\Delta f}{(1+d)} = \frac{100\text{k} - 50\text{k}}{1+1} =$ 25 Kbps

Problem6: We need to send data 3 bits at a time at a bit rate of 3 Mbps. The carrier frequency is 10 MHz. Calculate the **number of levels** (different frequencies), the **baud rate**, and the **bandwidth. Illustrate the frequency spectrum** showing the bandgap between the required carrier frequencies.

➔ **Is given,**

$r = 3$ bits

$N = 3Mbps$

$f_c = 10\ MHz$

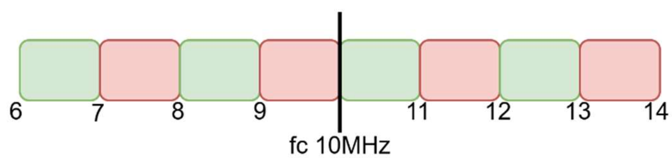
We Know,

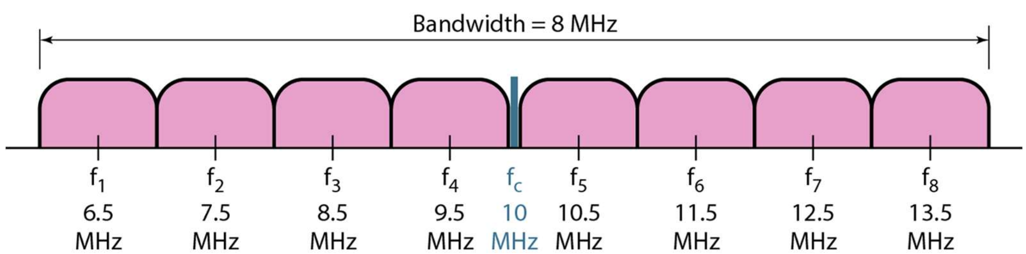
$$\boldsymbol{r = log_2 L}$$

$$\therefore \mathbf{L = 2^r} = 2^3 = 8$$

$$\therefore \boldsymbol{S = N \times \frac{1}{r}} = 3 \times \frac{1}{3} = 1Mbaud$$

$$\therefore \boldsymbol{B_{total} = L \times S} = 8 \times 1 = 8\text{ MHz}$$





Problem7: We need to send data 2 bits at a time at a bit rate of 4 Mbps. The carrier frequency is 50 MHz. Calculate the **number of levels** (different frequencies), the **baud rate**, and the **bandwidth. Illustrate the frequency spectrum** showing the bandgap between the required carrier frequencies.

➔ **Is given,**

$r = 2$ bits

$N = 4Mbps$

$f_c = 50\ MHz$

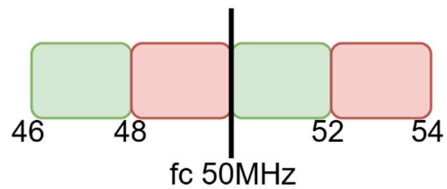
We Know,

$r = log_2 L$

$\therefore\ L = 2^r = 2^2 = 4$

$\therefore S = N \times \frac{1}{r} = 4 \times \frac{1}{2} = 2Mbaud$

$\therefore B_{total} = L \times S = 4 \times 2 = 8$ MHz



Problem8: We need to send data 4 bits at a time at a bit rate of 4 Mbps. The carrier frequency is 25 MHz. Calculate the **number of levels** (different frequencies), the **baud rate**, and the **bandwidth. Illustrate the frequency spectrum** showing the bandgap between the required carrier frequencies.

➔ **Is given,**

$r = 4$ bits

$N = 4Mbps$

$f_c = 25\ MHz$

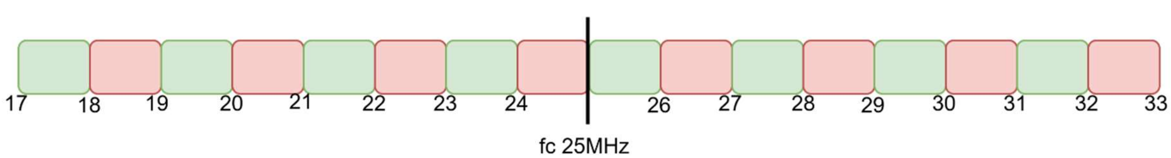
We Know,

$r = log_2 L$

$\therefore\ L = 2^r = 2^4 = 16$

$\therefore S = N \times \frac{1}{r} = 4 \times \frac{1}{4} = 1Mbaud$

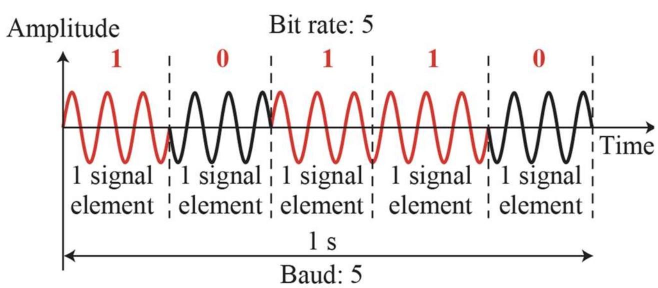
$\therefore B_{total} = L \times S = 16 \times 1 = 16$ MHz

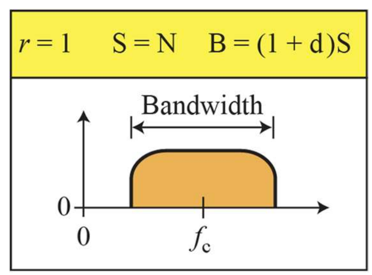


Binary Phase Shift Keying

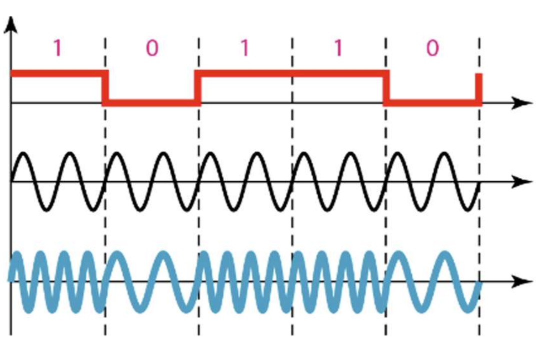
- In **PSK (Phase Shift Keying)**, **carrier phase** is changed to represent data.
- **Amplitude and frequency** stay **constant**.
- **PSK** is more common than **ASK** or **FSK**.
- **QAM** (combining ASK and PSK) is the most widely used digital-to-analog modulation today.

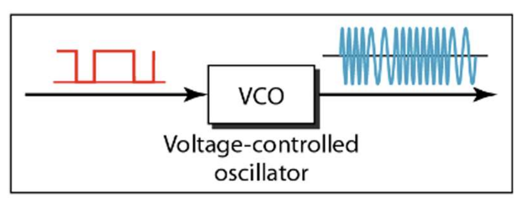
BPSK uses **two phases**: **0° and 180°** to represent binary data (1 and 0).





Implementation of binary BPSK





Formulas

Bandwidth

Bandwidth:	Max Bandwidth: (d=1)
$$\mathbf{B = (1 + d) \times S}$$ Where: • **B** = Bandwidth required (in Hz) • **S** = Signal (baud) rate • **d** = **Modulation factor** or **roll-off factor** (0~1)	$$\mathbf{B_{max} = 2S}$$
	Max Bandwidth: (d=0) $$\mathbf{B_{min} = S}$$

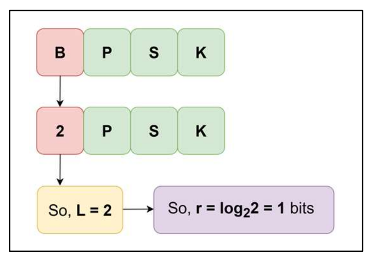
Problem9: A binary phase shift keying (BPSK) system is transmitting data at a bit rate of **10 kbps**.

1. What is the **baud rate** of the signal?
2. If the system uses a roll-off factor d = 0.5, calculate the **minimum bandwidth** required.

➔ **Is given,**

$r = 1$ bits

$N = 10kbps$



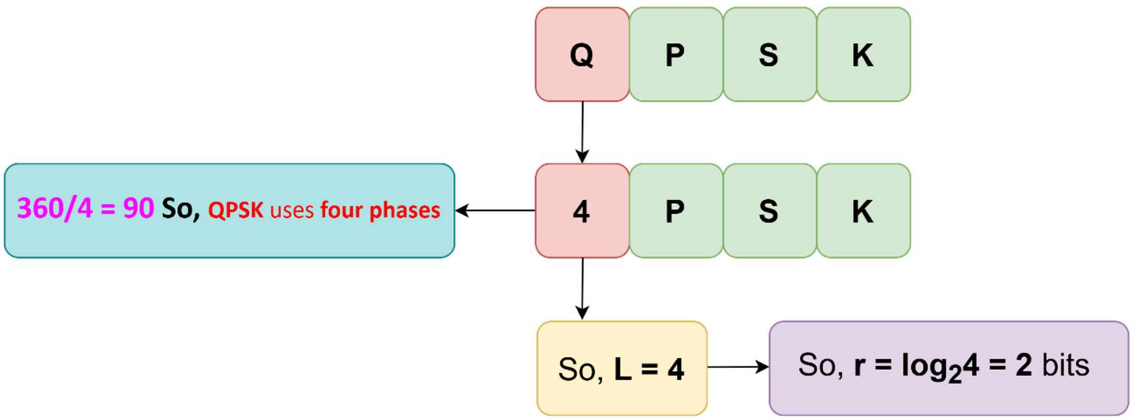
We Know,

$$\therefore S = N \times \frac{1}{r} = 10K \times \frac{1}{1} = 10Kbaud$$

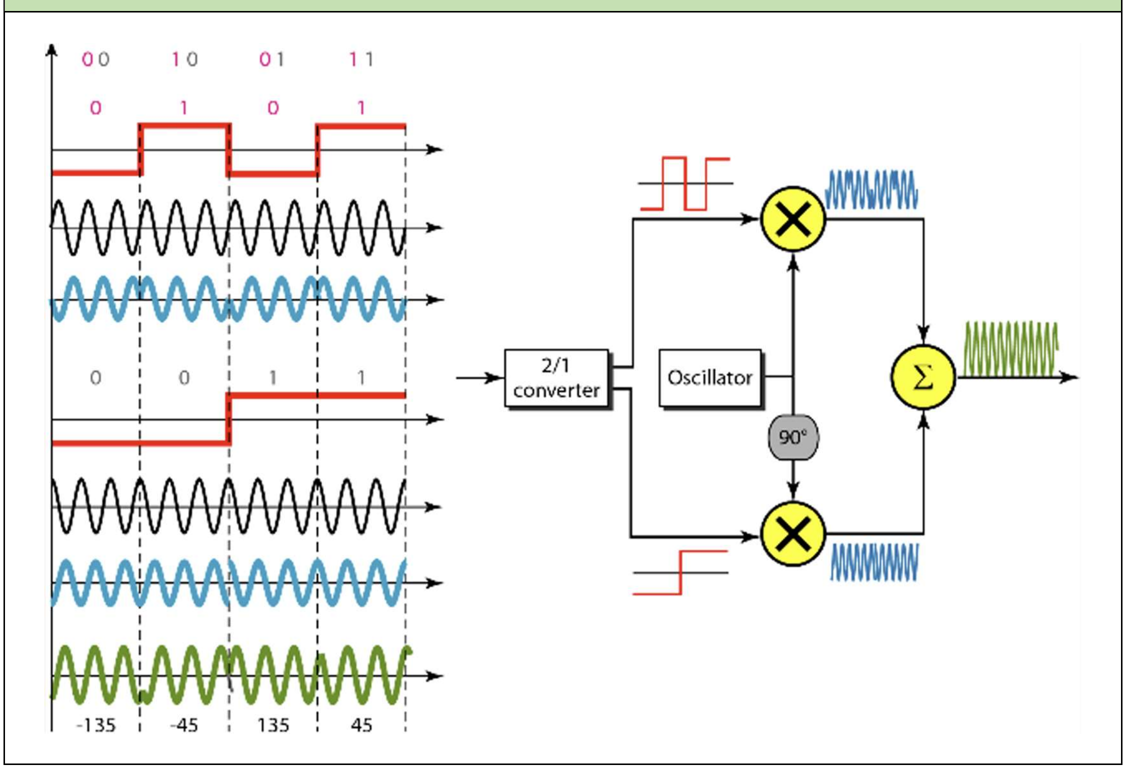
$$\therefore B_{min} = (1 + d) \times S = (1 + 0.5) \times 10\text{K} = 15\text{ KHz}$$

Quadrature PSK (QPSK)

- **QPSK** uses **2 bits per symbol**, reducing **baud rate** and **bandwidth**.
- It combines **two BPSK signals**: one **in-phase**, one **quadrature** (90° out of phase).
- Bits are split via **serial-to-parallel conversion** to two modulators.
- Each BPSK path processes bits at **half the original frequency** (bit duration becomes **2T**).



Implementation of binary BPSK



Problem10: Find the bandwidth for a signal transmitting at 12 Mbps for QPSK. The value of d = 0.

➔ **Is given,**

$L = 4$ [since QPSK]

$N = 12Mbps$

$d = 0$

We Know,

$\therefore \boldsymbol{r = log_2 L} = \log_2 \times 4 = 2 \text{ bits}$

$$\therefore \boldsymbol{S = N \times \frac{1}{r}} = 12M \times \frac{1}{2} = 6Mbaud$$

$\therefore \boldsymbol{B = (1 + d) \times S} = (1 + 0) \times 6\text{M} = 6 \text{ MHz}$

Problem11: A QPSK system transmits data at a **bit rate of 40 kbps**.

1. What is the **baud rate** of the signal?
2. If the roll-off factor d = 0.3, calculate the **minimum bandwidth** required.

➔ **Is given,**

$L = 4$ [since QPSK]

$N = 40kbps$

$d = 0.3$

We Know,

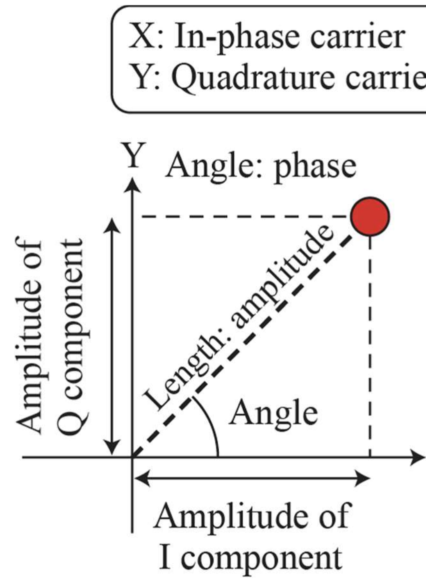
$\therefore \boldsymbol{r = log_2 L} = \log_2 \times 4 = 2 \text{ bits}$

$$\therefore \boldsymbol{S = N \times \frac{1}{r}} = 40k \times \frac{1}{2} = 20k\ baud$$

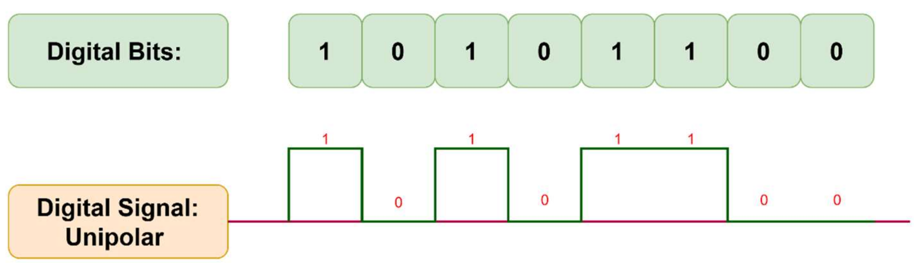
$\therefore \boldsymbol{B = (1 + d) \times S} = (1 + 0.3) \times 20\text{k} = 26 \text{ kHz}$

Constellation Diagram

- The diagram has **X (in-phase)** and **Y (quadrature)** axes.
- Each point represents a signal element with four key details:
    1. X-axis projection → Peak amplitude of in-**phase component**
    2. Y-axis projection → Peak amplitude of **quadrature component**
    3. Line length from origin → Total signal amplitude
    4. Line angle with X-axis → Signal phase



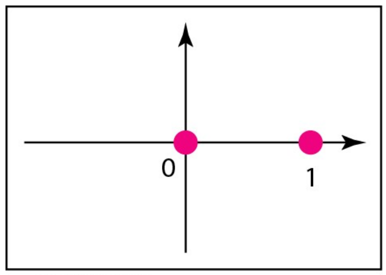
Constellation diagrams for ASK



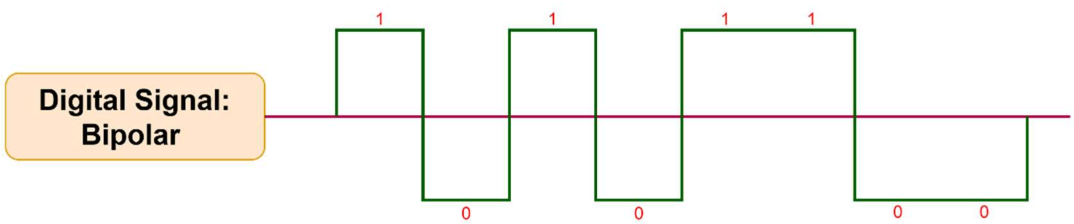
➢ **For ASK:**

"1" is represented by +V **volts.**

"0" is represented by **0 volts.**

➢ **ASK (Amplitude Shift Keying)** uses only the **in-phase (X-axis) carrier**. ➢ **Binary 0** → Amplitude = 0V → Point at **origin (0,0)** ➢ **Binary 1** → Amplitude = 1V → Point at **(1,0)** ➢ All signal points lie **on the X-axis** (no quadrature component).
 a. ASK (OOK)

Constellation diagrams for BPSK

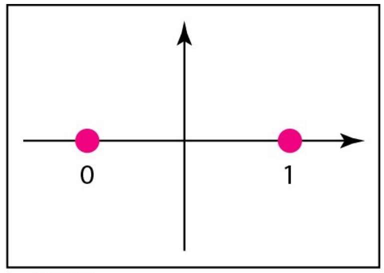


➢ **For BPSK:**

"1" is represented by +V **volts.**

"0" is represented by **-V volts.**

- **BPSK (Binary Phase Shift Keying)** uses only the **in-phase (X-axis) carrier**.
- Uses **polar NRZ** for modulation.
- Two signal elements:
  - **Binary 1** → Amplitude **+1V** (0° phase) → Point at **(+1, 0)**
  - **Binary 0** → Amplitude **−1V** (180° phase) → Point at **(−1, 0)**
- Points lie on the **X-axis**, with **opposite phases**.



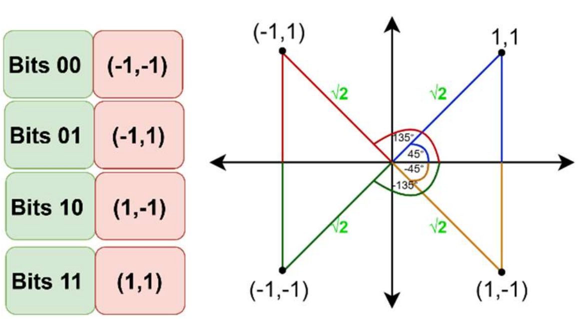
b. BPSK

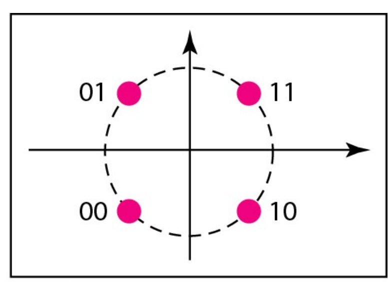
Constellation diagrams for QPSK

- Uses **two carriers**: **in-phase (X-axis)** and **quadrature (Y-axis)**.
- Each symbol represents **2 bits** **(e.g., `11`, `10`, `00`, `01`).**
- **For `11`:**
  - In-phase = 1V, Quadrature = 1V → Combined amplitude = √2
  - **Phase = 45°**
- All four signal points have amplitude **√2** and phases: **45°, 135°, −135°, −45°**.
- Amplitude can be normalized to **1V** by scaling carrier amplitude to **1/√2**.

"1" is represented by **+V volts.**

"0" is represented by **-V volts.**





c. QPSK

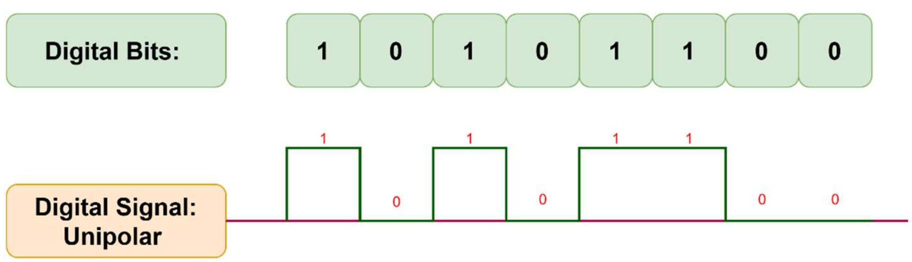
Quadrature Amplitude Modulation (QAM)

- **PSK** is limited by difficulty in detecting **small phase changes**, restricting bit rate.
- So far, only **one wave property** (amplitude, frequency, or phase) was altered at a time.
- **QAM (Quadrature Amplitude Modulation)** combines **ASK and PSK**.
- It uses **two carriers** (in-phase & quadrature) with **varying amplitudes** to represent more bits.

Constellation diagrams for some QAMs

Constellation diagrams for 4-QAM: Unipolar NRZ

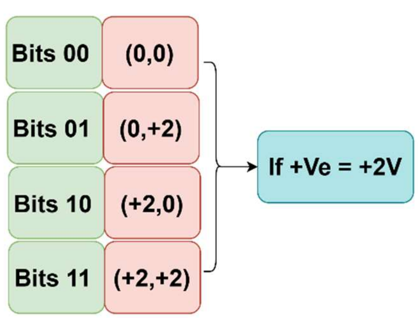
4-QAM: Uses **unipolar NRZ**, similar to **ASK (On-Off Keying)** for each carrier.

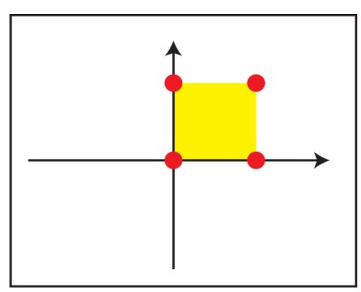


➢ **For 4-QAM: Unipolar NRZ**

"1" is represented by **+V volts.**

"0" is represented by **0 volts.**

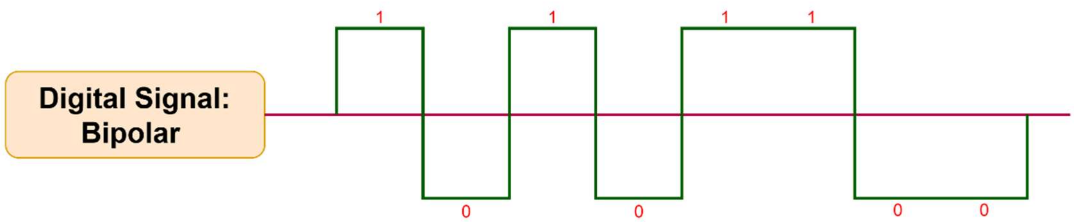




a. 4-QAM

Constellation diagrams for 4-QAM: Polar NRZ

4-QAM: Uses **polar NRZ**, which is **equivalent to QPSK.**



- **For 4-QAM: Polar NRZ**

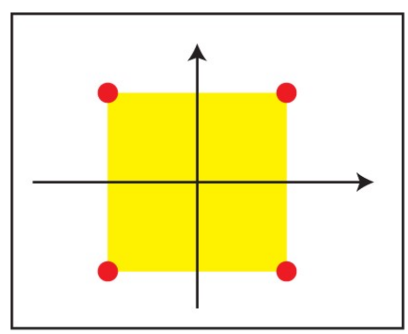
"1" is represented by **+V volts.**

"0" is represented by **-V volts.**

Bits 00	(-2,-2)
Bits 01	(-2,+2)
Bits 10	(+2,-2)
Bits 11	(+2,+2)

If +Ve = +2V

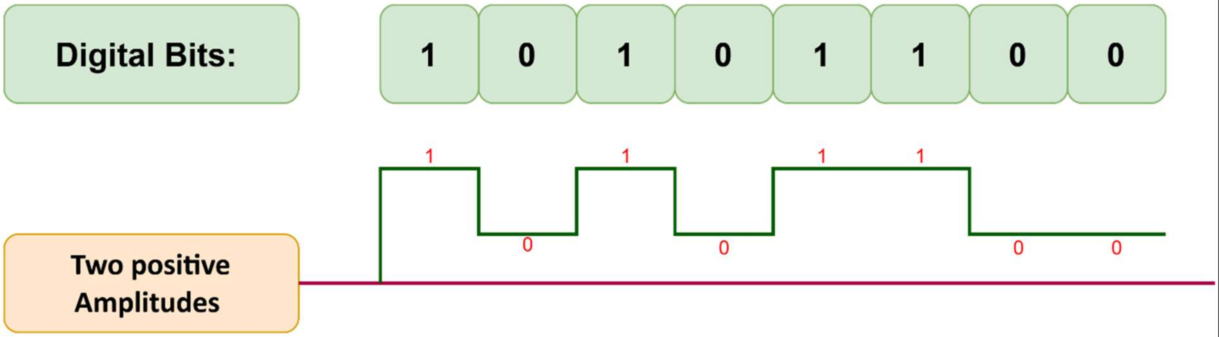
If -Ve = -2V



b. 4-QAM

Constellation diagrams for 4-QAM: Two positive amplitude

4-QAM: Uses **two positive amplitude levels** per carrier (not just on/off).



➢ **For 4-QAM: Two positive amplitude**

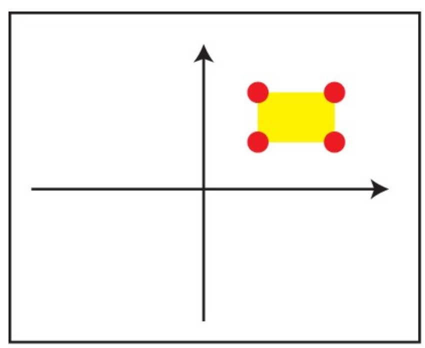
"1" is represented by +Ve volts.

"0" is represented by +Ve volts.

Bits 00	(+2,+2)
Bits 01	(+2,+4)
Bits 10	(+4,+2)
Bits 11	(+4,+4)

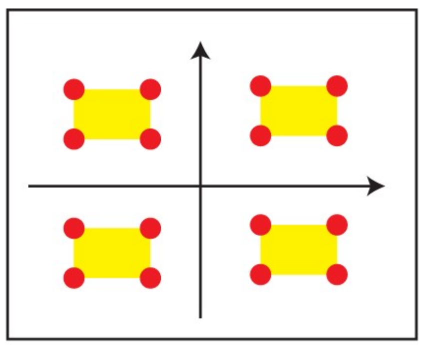
If "1" = +4V

If "0" = +2V



c. 4-QAM

Constellation diagrams for 16-QAM: Eight amplitude levels

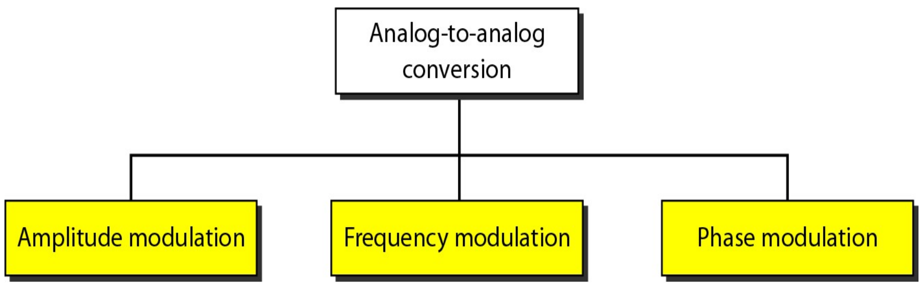
16-QAM: Uses **eight amplitude levels** total (4 positive, 4 negative) across in-phase and quadrature carriers.
➢ **For 16-QAM:** **"1"** is represented by **+Ve/-Ve/-Ve/+Ve volts.** **"0"** is represented by **+Ve/+Ve/-Ve/-Ve volts.**
 d. 16-QAM

ANALOG TRANSMISSION AND BANDWIDTH

Analog to Analog Conversion

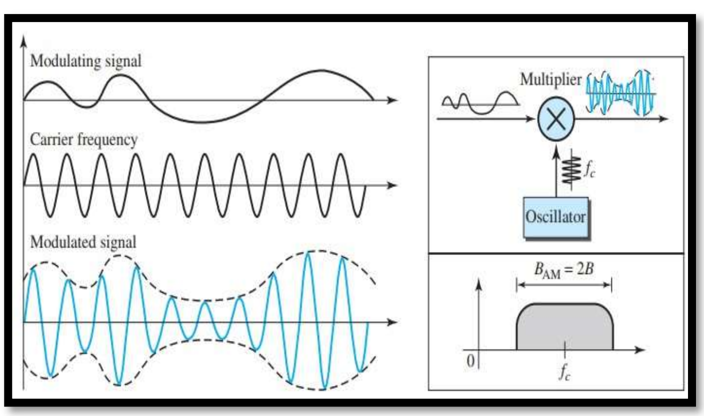
- **Analog-to-analog conversion (analog modulation)**: Represents **analog data** using an **analog carrier signal**.
- **Why modulate?**
    - Required when the **transmission medium is bandpass**.
    - Or when only a **bandpass channel** is available.

Analog to Analog Conversion Types



Amplitude Modulation (AM)

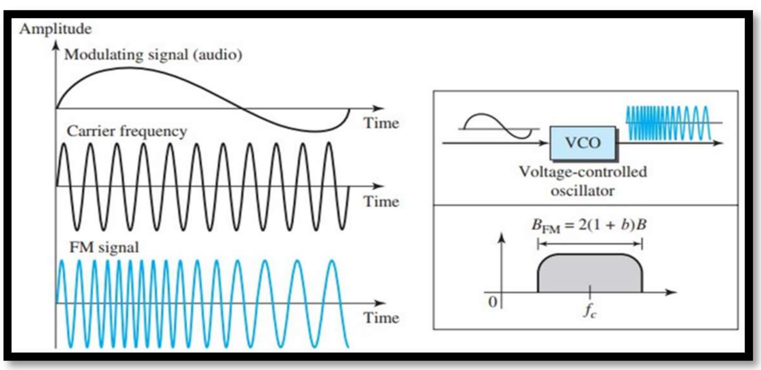
- **AM** changes the **amplitude** of the **carrier** based on the **modulating signal**.
- **Carrier's frequency and phase stay constant**.
- The **modulating signal forms the envelope** of the carrier.
- AM is implemented using a **multiplier circuit**.
- **Higher modulating amplitude → higher carrier amplitude**.
- The **envelope (dashed line)** represents the original information signal.



Formulas
Bandwidth
Bandwidth: $$\boxed{B_{AM} = (2B)}$$ Where: **B** = Bandwidth required (in Hz)

Frequency Modulation (FM)

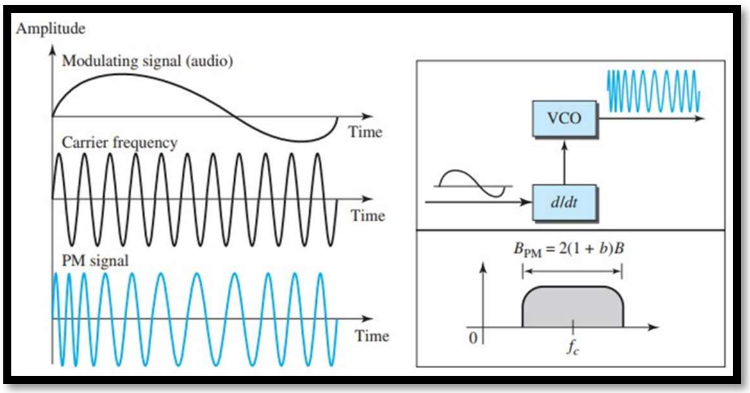
- **FM** varies the **frequency** of the **carrier** based on the **modulating signal's amplitude**.
- **Amplitude and phase** of the carrier remain **constant**.
- **Higher modulating amplitude → higher deviation in carrier frequency**.
- The **frequency** of the FM signal changes to **follow** the input signal's amplitude.



Formulas
Bandwidth
Bandwidth: $$\boxed{B_{FM} = 2(1 + b)B}$$ Where: **B** = Bandwidth required (in Hz) **b** = modulation-dependent factor, commonly **4**, used in **bandwidth calculations**.

Phase Modulation (PM)

- **PM** varies the **phase** of the **carrier** based on the **modulating signal's amplitude**.
- **Amplitude and frequency** of the carrier stay **constant**.
- **Greater modulating amplitude → greater phase shift** in the carrier.
- **PM vs. FM**:
  - **FM**: Frequency change ∝ **amplitude** of modulating signal
  - **PM**: Frequency change ∝ **derivative** of modulating signal



Formulas
Bandwidth
Bandwidth: $$\mathbf{B_{PM} = 2(1 + b)B}$$ Where: **B** = Bandwidth required (in Hz) **b** = modulation-dependent factor, commonly **4**, used in **bandwidth calculations**.

Bandwidth Utilization: Multiplexing

- **Multiplexing** enables **multiple signals** to be sent **simultaneously over one data link**.
- It's a solution to growing **traffic demands** in communication systems.
- Instead of adding more links, use **high-bandwidth links** to **carry multiple channels** efficiently.

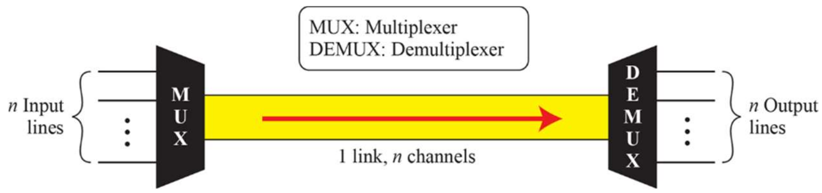
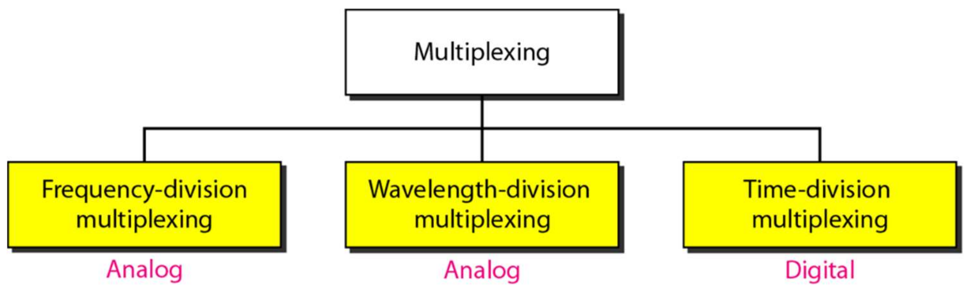


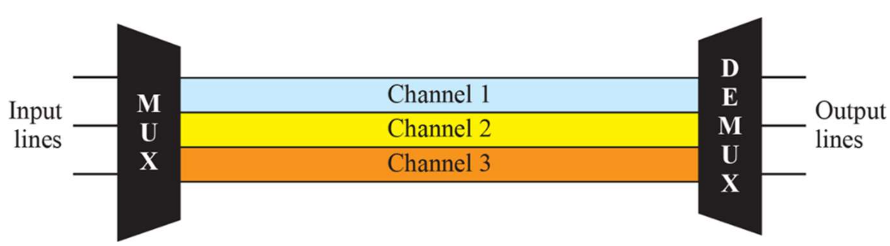
Fig: Dividing a link into channels

Types of multiplexing



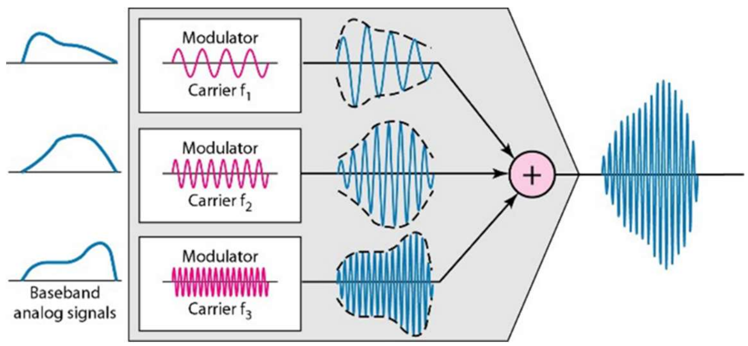
Frequency-Division Multiplexing (FDM)

- **FDM** is an **analog multiplexing** technique.
- Used when **link bandwidth > sum of all signal bandwidths**.
- Each signal **modulates a different carrier frequency**.
- Modulated signals are **combined into one composite signal** for transmission.



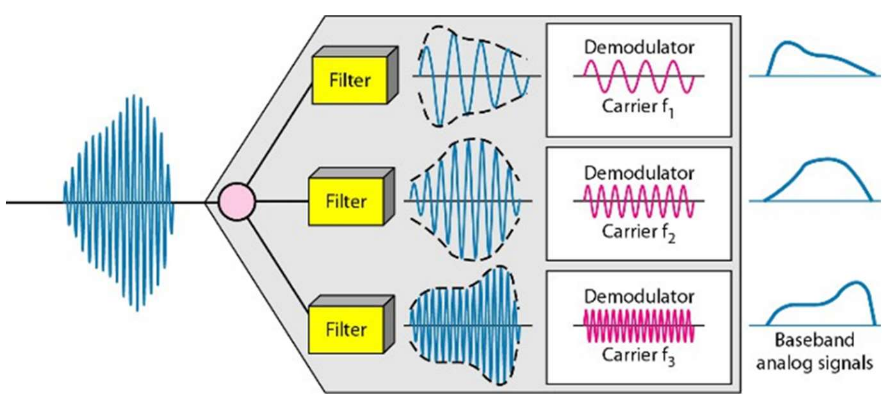
FDM Process

- Each source signal has a **similar frequency range**.
- In the **multiplexer**, each signal modulates a **different carrier frequency** (e.g., f_1, f_2, f_3).
- The **modulated signals are combined** into a **single composite signal**.
- Sent over a **high-bandwidth link** capable of carrying all combined signals.



FDM Demultiplexing

- The **demultiplexer uses filters** to **separate the composite signal** into individual components.
- Each component is sent to a **demodulator** to extract the **original signal** from its carrier.
- The extracted signals are then sent to their **respective output lines**.



Problem1: Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

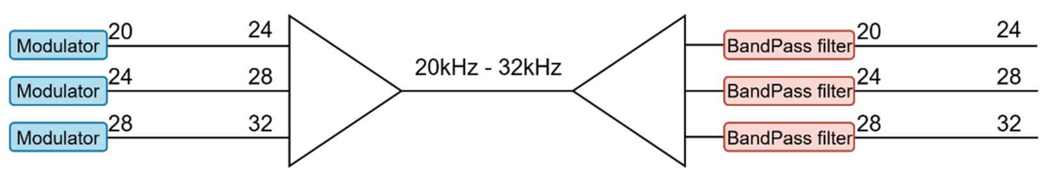
➔ **Is given,**

- Each **voice channel** = 4 kHz bandwidth
- Total **link bandwidth** = 12 kHz (from 20 kHz to 32 kHz)
- No guard bands

Frequency Allocation:

Channel	Frequency Range (kHz)
Ch 1	20 – 24
Ch 2	24 – 28
Ch 3	28 – 32

Frequency Domain Representation:



Guard Bands (GB)

⇨ **Guard bands** are small frequency gaps placed between adjacent channels in **Frequency-Division Multiplexing (FDM)** to:

- **Without guard bands**: All channels are packed tightly.
- **With guard bands**: Extra frequency space is used to improve signal separation and reduce interference.

If **guard bands are included**, the 12 kHz spectrum (20–32 kHz) must be split **not just into 3×4 kHz**, but with **space between channels**.

For example, if you use 1 kHz guard bands, the configuration might look like:

Channel	Frequency Range (kHz)
Ch 1	20 – 24
Guard	24 – 25
Ch 2	25 – 29
Guard	29 – 30
Ch 3	30 – 34

➡ **But notice:** with guard bands, we now need **more than 12 kHz** total bandwidth.

Formulas

Total Bandwidth

Total Bandwidth:

$$\boxed{B_{total} = No\ of\ ch \times f_{each\ ch} + No\ of\ GB \times f_{each\ GB}}$$

$$\boxed{No\ of\ GB = No\ of\ ch - 1}$$

Where:

B = Bandwidth required (in Hz)

$f_{each\ ch}$ = frequency for each channel

$f_{each\ GB}$ = frequency for each Guard Band

Problem2: Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

➔ **Is given,**

- Each **channel** = 100 kHz bandwidth
- Each Guard Band = 10 kHz
- No of $ch = 5$

We Know,

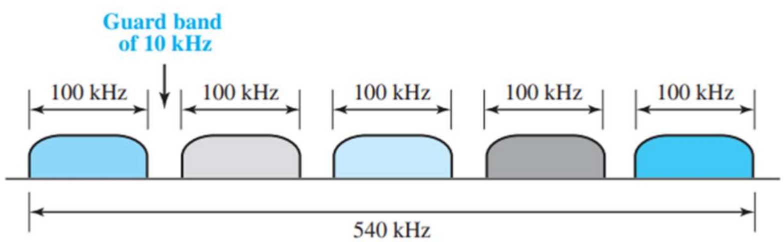
$$\therefore \boldsymbol{no\ of\ GB = no\ of\ ch - 1} = 5 - 1 = 4$$

$$\therefore \boldsymbol{B_{total} = No\ of\ ch \times f_{each\ ch} + No\ of\ GB \times f_{each\ GB}}$$

$$= 5 \times 100K + 4 \times 10k$$

$$= 500K + 40k$$

$$= 540\ kHz$$



Problem3: Four data channels (digital), each transmitting at 1 Mbps, use a satellite channel of 1 MHz. Design an appropriate configuration, using FDM

➔ **Is given,**

- **No. of channels = 4**
- **Digital data rate = 1Mbps**
- B_{total} = 1MHz

Since, FDM is an **analog technique**, we need that **each digital channel is modulated into an analog signal** using **analog modulation**. For that we need modulation scheme

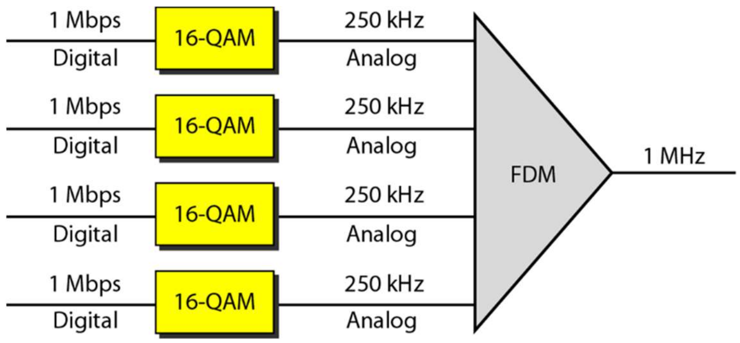
So, Per channel Bandwidth:

$$B_{pch} = \frac{B_{total}}{No.of\ ch} = \frac{1MHz}{4} = 250kHz$$

$$\text{order "M"} = 2^{\frac{Data\ Rate}{B_{pch}}} = 2^{\frac{1Mb}{250kHz}} = 2^4 = 16$$

∴ So, required modulation scheme = 16QAM

Frequency Domain Representation:

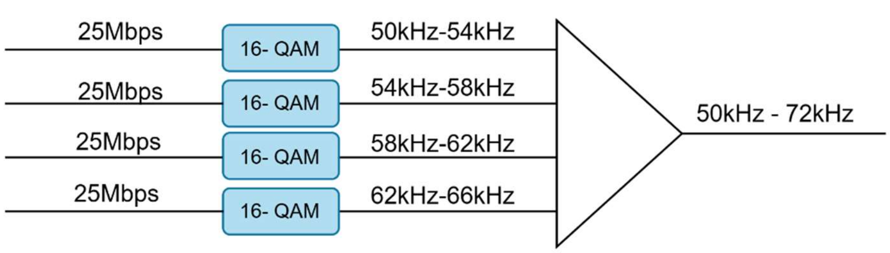


Problem3: You are given four digital data channels, each transmitting at **25 Mbps**. Each channel, when modulated, occupies a bandwidth of **4 kHz**. The goal is to combine these four channels using **Frequency Division Multiplexing (FDM)** into a link with an overall bandwidth of **28 kHz**, spanning the frequency range from **50 kHz to 78 kHz**. A **guard band of 2 kHz** is required between adjacent channels to minimize interference.
Design an appropriate FDM configuration and compute the **total bandwidth consumed by guard bands**. Also, present a frequency-domain layout of the configuration.

➔ **Is given,**

- **Number of data channels = 4**
- **Bandwidth per channel = 4 kHz**
- **Guard band between channels = 2 kHz**
- **Total available bandwidth = 28 kHz**
- **Frequency range = 50 kHz to 78 kHz**

Since, FDM is an **analog technique**, we need that **each digital channel is modulated into an analog signal** using **analog modulation**.

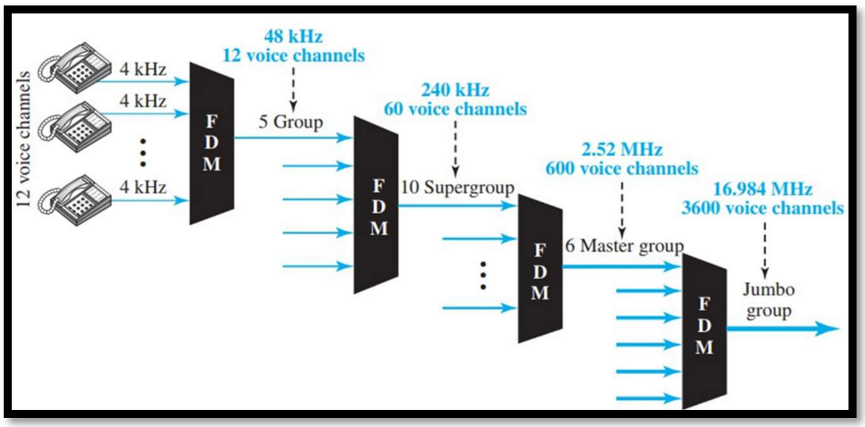


$$\therefore no\ of\ GB = no\ of\ ch - 1 = 4 - 1 = 3$$

$$\therefore GB_{total} = No\ of\ GB\ \times f_{each\ GB} = 3 \times 2 = 6 \text{ kHz}$$

Analog Hierarchy

- Telephone companies use **multiplexing** to **combine multiple low-bandwidth lines** into **fewer high-bandwidth lines**.
- For **analog lines**, they use **Frequency-Division Multiplexing (FDM)**.
- The system is organized hierarchically into **groups**, **supergroups**, **master groups**, and **jumbo groups**.



> - $1G = 12VC$
> - $1SG = 5G$
> - $1MG = 10SG$
> - $1JG = 6MG$

- **Group**: Combines **12 voice channels** → **48 kHz** bandwidth
- **Supergroup**: Combines **5 groups or 60 voice channels** → **240 kHz**
- **Master group**: Combines **10 supergroups (600 voice channels)**
  - Requires **2.40 MHz**, but with guard bands uses **2.52 MHz**
- **Jumbo group**: Combines **6 master groups**
  - Needs **15.12 MHz**, extended to **16.984 MHz** with guard bands

BANDWIDTH UTILIZATION

Wavelength-division multiplexing (WDM)

- **WDM** uses fiber-optic cable's high data rate efficiently.
- A single data line on fiber wastes bandwidth; WDM combines multiple lines.
- Conceptually similar to **FDM** but uses **optical signals**.
- Combines multiple **high-frequency light signals** into one fiber.
- At the receiver, a **demultiplexer** separates the signals.
- Figure 6.10 illustrates the WDM system.

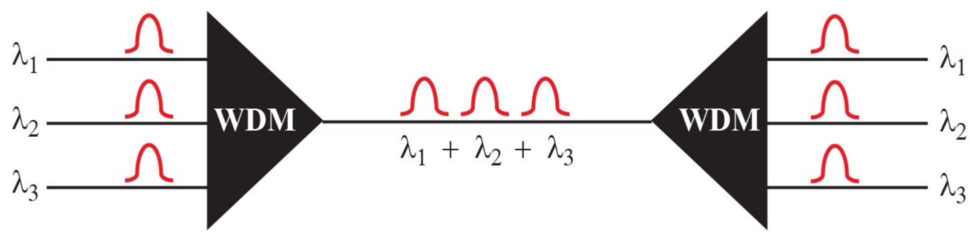


Fig6.10: Wavelength-division multiplexing

- WDM is **technically complex** but based on a **simple idea**.
- It **combines multiple light sources** into one at the **multiplexer**.
- It **separates them** at the **demultiplexer**.
- **Prisms** can be used to combine and split the light signals.

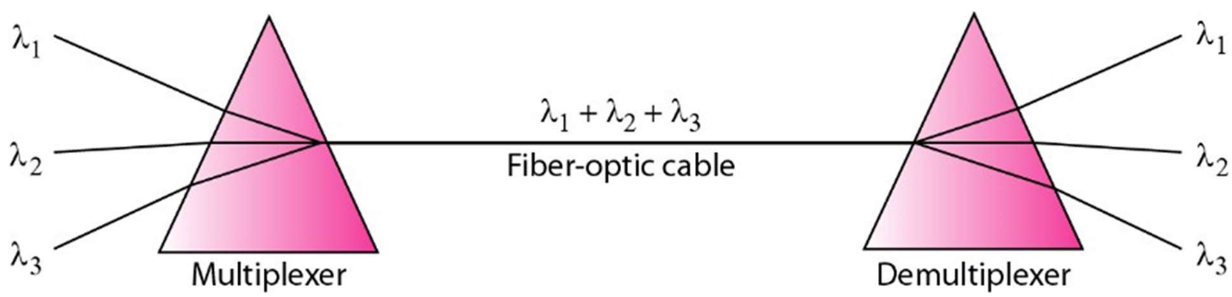


Fig6.11: Prisms in wave-length division multiplexing

Time-division Multiplexing (TDM)

- **TDM** is a **digital multiplexing** technique.
- It **shares time**, not bandwidth like FDM.
- Each connection gets a **time slot** on the link.
- The **same link** is used, but divided by **time** instead of frequency.
- Signals **1, 2, 3, & 4** are sent **sequentially** over time.

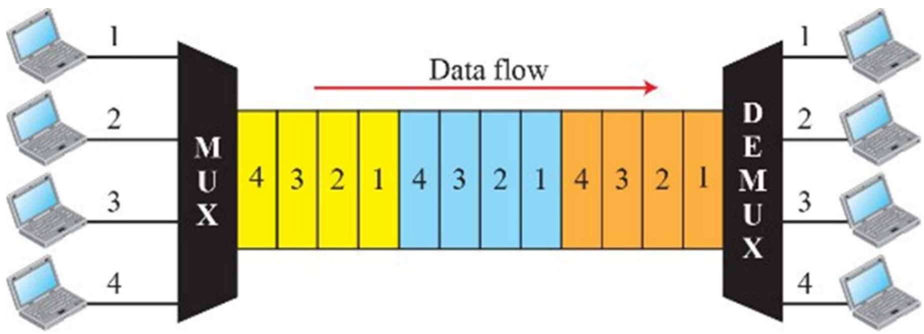


Fig6.12: TDM

Synchronous TDM

- **Two types of TDM**: **Synchronous** and **Statistical**.
- In **Synchronous TDM**, each connection gets a **fixed time slot**, even if it's idle.
- **Time slots and frames**:
    - Each input has a **time slot** in each frame.
    - Output time slots are **shorter**: $\frac{1}{n}$ of the input slot duration (n = number of inputs).

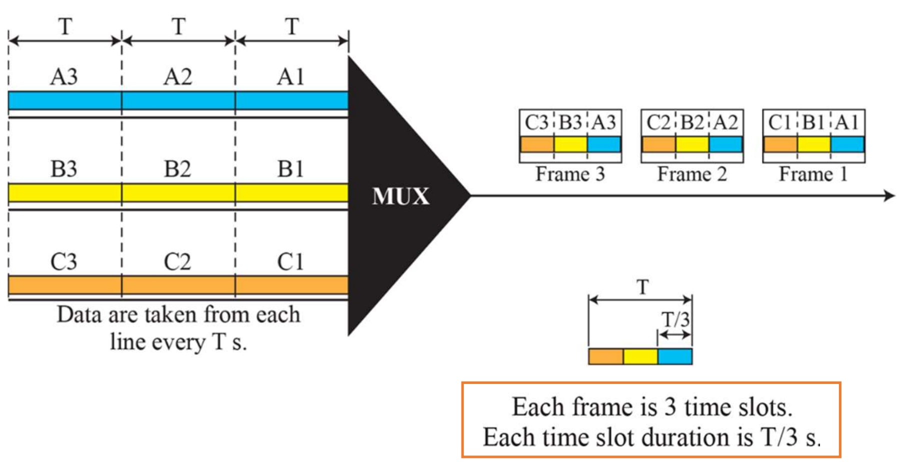


Fig6.13: Synchronous time-division multiplexing

Formulas

Input Bit Duration

$$\text{Input Bit Duration} = \frac{1}{\text{Input Data Rate}}$$

Input Slot Duration

$$\text{Input Slot Duration} = \frac{1}{\text{Input Data Rate}}$$

Output Bit Duration

$$\text{Output Bit Duration} = \frac{1}{\text{Output Data Rate}}$$

Output Slot Duration

$$\text{Output Slot Duration} = \frac{\text{Input Slot Duration}}{no\ of\ input\ lines}$$

Output Data Rate

$$\text{Output Data Rate} = \text{Frame Size} \times \text{Frame Rate}$$

Or

$$\text{Output Data Rate} = \text{Input Data Rate} \times No\ of\ input\ lines$$

Frame Rate

$$\text{Frame Rate} = \frac{\text{Input Data Rate (per line)}}{\text{Number of Bits per Time Slot}}$$

Frame Duration

$$\text{Frame Duration} = \frac{1}{\text{Frame Rate}}$$

Frame Size (in bits)

$$\text{Frame Size} = \text{Number of Input Lines} \times \text{Number of Bits per Time Slot}$$

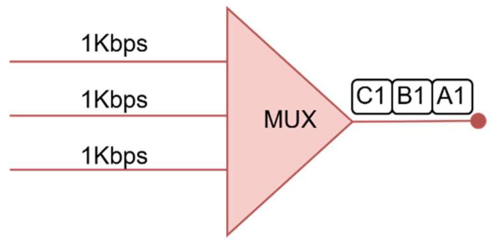
Problem1: Figure shows synchronous TDM with data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed, what is the duration of

a. Each input slot,

b. Each output slot, and

c. Each frame,

d. Output bit/Data Rate,



➔ **Is given,**

- **Input data rate = 1Kbps**
- **Number of Bits per Time Slot = 1 bit**
- **n = input lines = 3**

We Know,

(a)

$$\text{Input Slot Duration} = \frac{1}{\text{Input Data Rate}} = \frac{1}{1kbps} = 1ms$$

(b)

$$\text{Output Slot Duration} = \frac{\text{Input Slot Duration}}{n = input\ lines} = \frac{1ms}{3} = \frac{1}{3}ms$$

(c)

$$\text{Frame Rate} = \frac{\text{Input Data Rate (per line)}}{\text{Number of Bits per Time Slot}} = \frac{1kbps}{1} = 1kfps$$

$$\text{Frame Duration} = \frac{1}{\text{Frame Rate}} = \frac{1}{1kbps} = 1ms$$

(d)

$$\textbf{Output Data Rate} = \textbf{Frame Size} \times \textbf{Frame Rate}$$
$$= (\textbf{Number of Input Lines} \times \textbf{Number of Bits per Time Slot}) \times 1kbps$$
$$= (3 \times 1) \times 1kbps = 3kbps$$

Or

$$\textbf{Output Data Rate} = \textbf{Data rate for each input connection} \times \textbf{No. of input lines}$$
$$= 1\text{kbps} \times 3 = 3\text{kbps}$$

Problem2: Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find
(a) the input bit duration,
(b) the output bit duration,
(c) the output bit rate, and
(d) the output frame rate.
(e) the output frame duration.

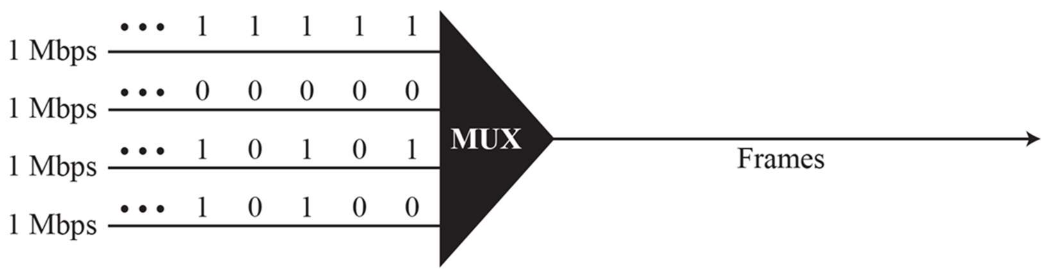


Fig6.14: Synchronous time-division multiplexing

➔ **Is given,**

- **Input data rate = 1Mbps**
- **Number of Bits per Time Slot = 1 bit**
- **n = input lines = 4**

We Know,

(a)

$$\textbf{Input bit Duration} = \frac{1}{\textbf{Input Data Rate}} = \frac{1}{1Mbps} = 1\mu s$$

(b)

$$\text{Output Data Rate} = \text{Data rate for each input connection} \times \text{No. of input lines}$$
$$= 1\text{Mbps} \times 4 = 4\text{Mbps}$$

$$\text{Output Bit Duration} = \frac{1}{\text{Output Data Rate}} = \frac{1}{4Mbps} = 250ns$$

(c)

$$\text{Output Data Rate} = \text{Data rate for each input connection} \times \text{No. of input lines}$$
$$= 1\text{Mbps} \times 4 = 4\text{Mbps}$$

(d)

$$\text{Frame Rate} = \frac{\text{Input Data Rate (per line)}}{\text{Number of Bits per Time Slot}} = \frac{1Mbps}{1} = 10^6 fps$$

(e)

$$\text{Frame Duration} = \frac{1}{\text{Frame Rate}} = \frac{1}{10^6} = 1\mu s$$

Problem3: Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find
(1) the duration of 1 bit before multiplexing,
(2) the transmission rate of the link,
(3) the duration of a time slot, and
(4) the duration of a frame.

➔ **Is given,**

- **Input data rate = 1kbps**
- **Number of Bits per Time Slot = 1 bit**
- **n = input lines = 4**

We Know,

(a)

$$\text{Input bit Duration} = \frac{1}{\text{Input Data Rate}} = \frac{1}{1kbps} = 1ms$$

(b)

$$\text{Output Data Rate} = \text{Data rate for each input connection} \times \text{No. of input lines}$$
$$= 1\text{kbps} \times 4 = 4\text{kbps}$$

(c)

$$\text{Output Slot Duration} = \frac{\text{Input Slot Duration}}{n = input\ lines} = \frac{1ms}{4} = 0.25ms = 250\mu s$$

(d)

$$\text{Frame Rate} = \frac{\text{Input Data Rate (per line)}}{\text{Number of Bits per Time Slot}} = \frac{1kbps}{1} = 10^3 fps$$

$$\text{Frame Duration} = \frac{1}{\text{Frame Rate}} = \frac{1}{10^3} = 1ms$$

Problem4: Four channels are multiplexed using TDM. If each channel sends 100 bytes /s and we multiplex 1 byte per channel, show

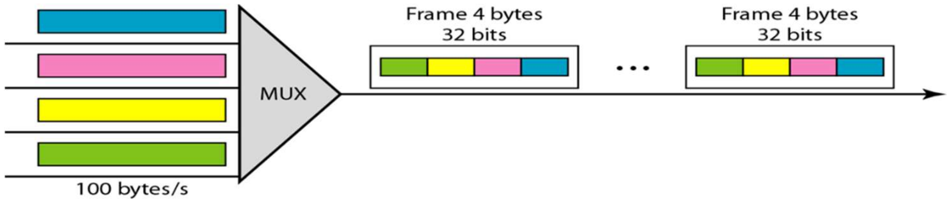
a) the frame traveling on the link,
b) the size of the frame,
c) the frame rate, and
d) the duration of a frame,
e) the bit rate for the link

➔ **Is given,**

- **Input data rate = 100 bytes /s**
- **Number of Bits per Time Slot = 1 byte per channel**
- **n = input lines = 4 channels**

We Know,

(a)



(b)

$$\text{Frame Size} = \text{Number of Input Lines} \times \text{Number of Bits per Time Slot}$$
$$= 4 \times 1 \times 8 = 32 \text{ bits}$$

(c)

$$\text{Frame Rate} = \frac{\text{Input Data Rate (per line)}}{\text{Number of Bits per Time Slot}} = \frac{100\ bytes/s}{1\ byte} = 100fps$$

(d)

$$\text{Frame Duration} = \frac{1}{\text{Frame Rate}} = \frac{1}{100} = 10ms$$

(e)

$$\text{Output Bit Rate} = \text{Data rate for each input connection} \times \text{No. of input lines}$$
$$= (100 \text{ bytes/s}) \times 4 \text{ byte} = 400 \text{ bytes} \times 8 = 3.2\text{kbps}$$

Problem5: A multiplexer combines four 100-kbps channels using a time slot of 2 bits.

a) Show the output with four arbitrary inputs.

b) What is the frame rate?

c) What is the frame duration?

d) What is the bit rate?

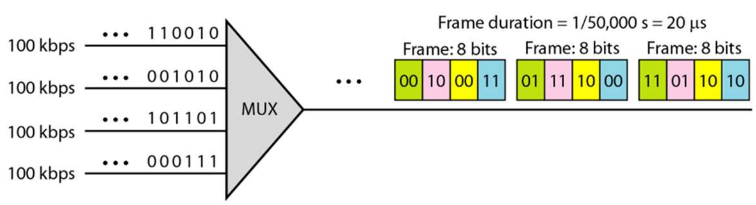
e) What is the bit duration?

➔ **Is given,**

- **Input data rate = 100 kbps**
- **Number of Bits per Time Slot = 2 bits**
- **n = input lines = 4**

We Know,

(a)



(b)

$$\textbf{Frame Rate} = \frac{\textbf{Input Data Rate (per line)}}{\textbf{Number of Bits per Time Slot}} = \frac{100kbps}{2} = 50k\ fps$$

(c)

$$\textbf{Frame Duration} = \frac{1}{\textbf{Frame Rate}} = \frac{1}{50k} = 20\mu s$$

(d)

$$\textbf{Output Bit Rate} = \textbf{Data rate for each input connection} \times \textbf{No. of input lines}$$

$$= 100\text{k} \times 4 = 400\ \text{kbps}$$

(e)

$$\textbf{Output Bit Duration} = \frac{1}{\textbf{Output Data Rate}} = \frac{1}{400kbps} = 2.5\mu s$$

Empty Slot

- Input links may occasionally have no data to send.
- This leads to unused slots on the output link.
- Causes bandwidth wastage.
- **Statistical TDM** improves efficiency by allocating slots dynamically.

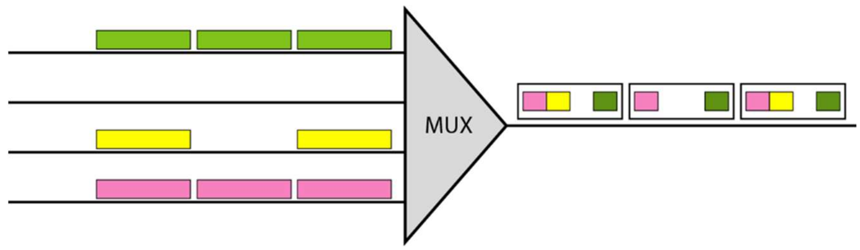
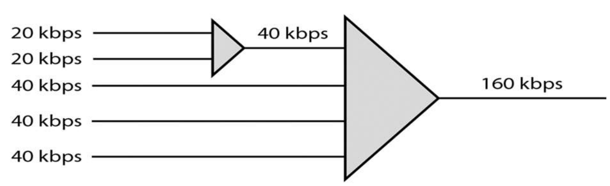


Fig20: Empty Slots.

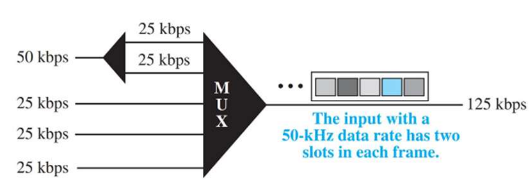
Data Rate Management

- **Issue**: Input links may have different data rates (some slower/faster).
- **Solution**: Use one of three strategies to match rates:
  - **Multilevel**: For input rates that are multiples of each other.
  - **Multislot**: When a GCD exists; assign more slots to higher-rate links.
  - **Pulse Stuffing**: When no GCD; insert extra bits to match slower links to faster ones.

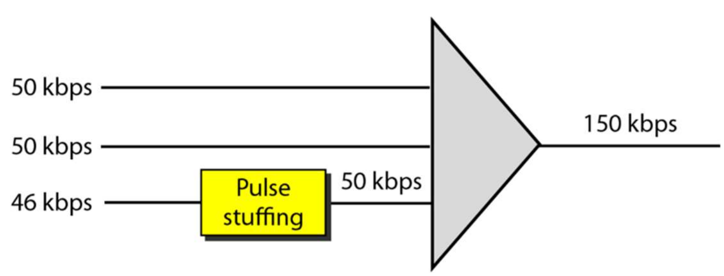
Multilevel



Multislot



Pulse Stuffing



Frame Synchronization

- **Synchronization** between multiplexer and demultiplexer is crucial.
- Without it, data may go to the wrong channel.
- **Framing bits** are added at the start of each frame for sync.
- These bits follow a known pattern (e.g., alternating 0 and 1).
- Helps demultiplexer align and separate time slots correctly.

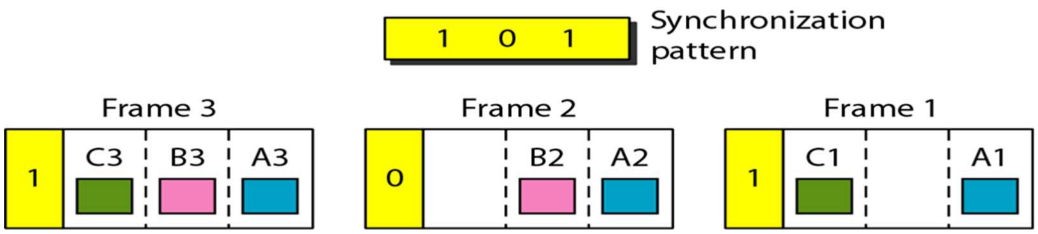


Fig24: Framing bits.

Problem6: We have four sources, each creating 250 8-bit characters per second. If the interleaved unit is a character and 1 synchronizing bit is added to each frame, find
(a) the data rate of each source,
(b) the duration of each character in each source,
(c) the frame rate,
(d) the duration of each frame,
(e) the number of bits in each frame, and
(f) the data rate of the link.

➔ **Is given,**

- **No of bits = 8 bits**
- **Each source sends 250 characters/sec**
- **No of input lines = 4**

We Know,

(a)

$$\textbf{Input Data Rate} = 250 \times 8 = 2\text{kbps}$$

(b)

$$\text{Duration for each character} = \frac{1}{Each\ characters/sec} = \frac{1}{250} = 4ms$$

(c)

$$\text{Frame Rate} = \frac{Input\ data\ rate}{no\ of\ bits} = \frac{2kbps}{8} = 250\text{fps}$$

(d)

$$\text{Frame Duration} = \frac{1}{\text{Frame Rate}} = \frac{1}{250} = 4ms$$

(e)

Each frame contains:

- 1 character from each of 4 sources
- Plus 1 **synchronization bit**

$$\textbf{Frame size} = (\textbf{No of input lines} \times \textbf{No of bits}) + \mathbf{1} = (\mathbf{4} \times \mathbf{8}) + \mathbf{1} = \mathbf{33\ bits}$$

(f)

$$\textbf{Output Data rate} = \textbf{Frame size} \times \textbf{Frame rate} = \mathbf{33} \times \mathbf{250} = \mathbf{8250\ bps}$$

Problem7: Two channels, one with a bit rate of 100 kbps and another with a bit rate of 200 kbps, are to be multiplexed.

a) How to solve this data rate mismatch problem?
b) What is the frame rate?
c) What is the frame duration?
d) What is the bit rate of the link?

➔ **Is given,**

- ➢ **Channel 1: 100 kbps**
- ➢ **Channel 2: 200 kbps**

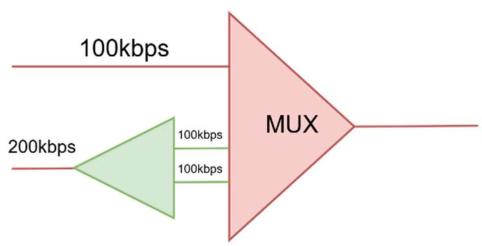
We Know,

(a)

Use **multislot TDM**, since one channel (200 kbps) is **twice** the speed of the other (100 kbps). This means:

- Allocate **1 time slot per frame** to the **100-kbps** channel
- Allocate **2 time slots per frame** to the **200-kbps** channel

So, total: **3 slots per frame**



(b)

$$\text{Frame Rate} = \frac{\text{Input Data Rate (per line)}}{\text{Number of Bits per Time Slot}} = \frac{100kbps}{1} = 100k\ fps$$

(c)

$$\text{Frame Duration} = \frac{1}{\text{Frame Rate}} = \frac{1}{100k} = 10\mu s$$

(d)

$$\text{Output Data Rate} = \text{Data rate for each input connection} \times \text{No. of input lines}$$
$$= 100\text{kbps} \times 3 = 300\text{kbps}$$

Digital Hierarchy

- **TDM (Time Division Multiplexing)** is used by telephone companies to transmit multiple signals over one medium.
- A **digital hierarchy** defines levels of digital signals, known as **DS (Digital Signal) levels**.
- Each level supports a specific **data rate** and **number of voice channels**.
- **DS0**: Basic level, 64 kbps – carries 1 voice channel.
- **DS1**: 1.544 Mbps – carries 24 voice channels.
- **DS2**: 6.312 Mbps – carries 96 voice channels.
- **DS3**: 44.736 Mbps – carries 672 voice channels.
- Higher levels combine multiple lower-level signals.

DS0 = 1VC
DS1 = 24 DS0
DS2 = 4 DS1
DS3 = 7 DS2
DS4 = 6 DS3

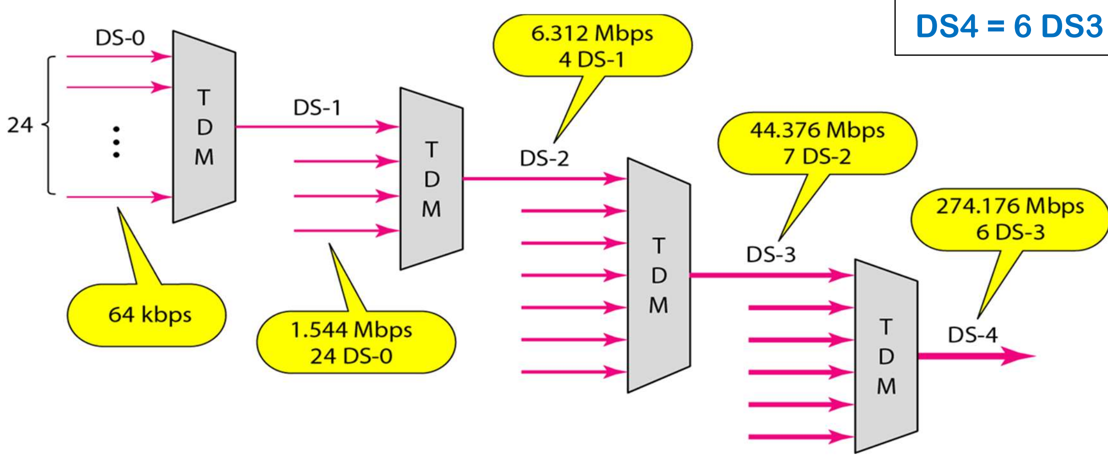


Fig25: Digital Hierarchy.

T Lines

- **DS-0, DS-1, etc.** are digital signal service levels.
- **T-lines (T-1 to T-4)** are physical transmission lines used to implement DS services.
- Each T-line's capacity matches the corresponding **DS level**:
  - **T-1** → DS-1 (1.544 Mbps)
  - **T-3** → DS-3 (44.736 Mbps)
- **T-1 and T-3** are the only ones **commercially available**.
- Used by telephone companies for high-capacity communication.

Service	Line	Rate (Mbps)	Voice Channels
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

T Lines for Analog Transmission

- **T lines** are designed for **digital transmission** (data, audio, video).
- They can also carry **analog signals**, like regular telephone calls.
- To transmit analog signals, they must be:
  - **Sampled**
  - **Converted to digital**
  - **Time-Division Multiplexed (TDM)**

This process allows analog signals to be sent over digital T lines.

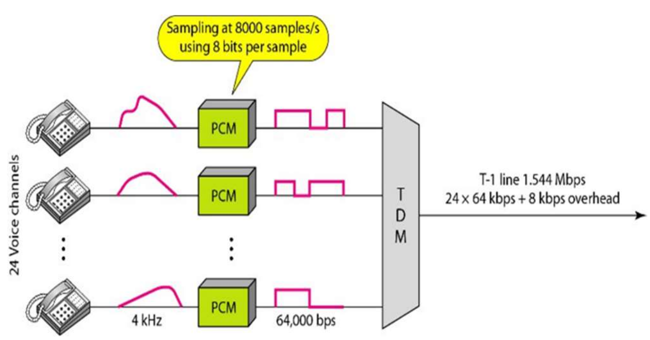


Fig26: T-1 line for multiplexing telephone lines.

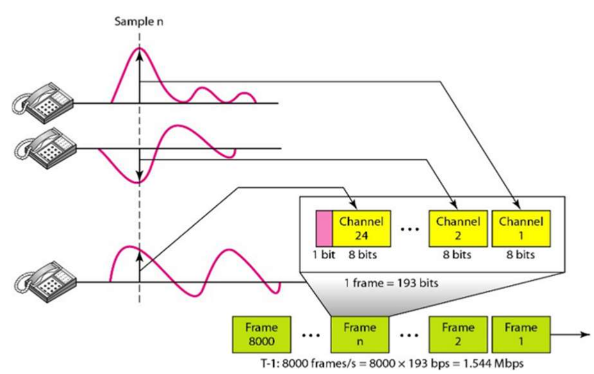


Fig27: T-1 Frame Structure.

E line rates

- **E lines** are the European equivalent of **T lines**.
- Conceptually similar to T lines (used for digital transmission).
- **Differences lie in data rates and capacities**.

Used in Europe for digital telecommunication systems.

Line	*Rate (Mbps)*	*Voice Channels*
E-1	2.048	30
E-2	8.448	120
E-3	34.368	480
E-4	139.264	1920

Statistical Time-Division Multiplexing

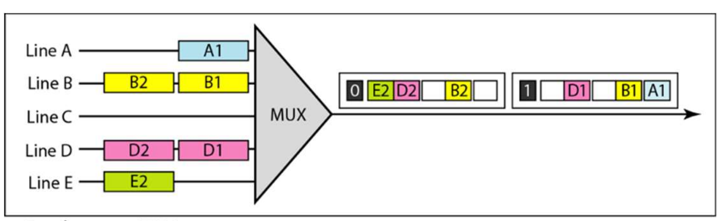
Synchronous TDM:

- Each input has a **reserved slot**, even if there's **no data** → **inefficient**.
- **No addressing needed** – fixed slot-input relationship.
- Output slot carries **only data**.
- Works if multiplexer and demultiplexer are **synchronized**.

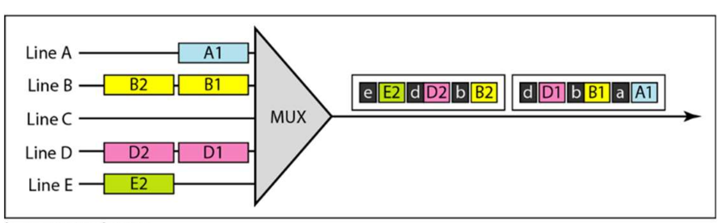
Statistical TDM (StatMux):

- **Slots are dynamically allocated** based on active input lines.
- **Fewer slots** than input lines → **more efficient**.
- **Round-robin** check: if line has data → allocate slot; else → skip.
- **Addressing required**: each slot includes data + **destination address**.
- No fixed slot-input relationship → requires addressing for correct delivery.

TDM slot comparison



a. Synchronous TDM



b. Statistical TDM

TRANSMISSION MEDIA & OPTICAL COMMUNICATION

Transmission Medium

- Transmission media are below the physical layer.
- They are directly controlled by the physical layer.
- Can be referred to as "Layer 0".
- Figure 7.1 shows their position under the physical layer.

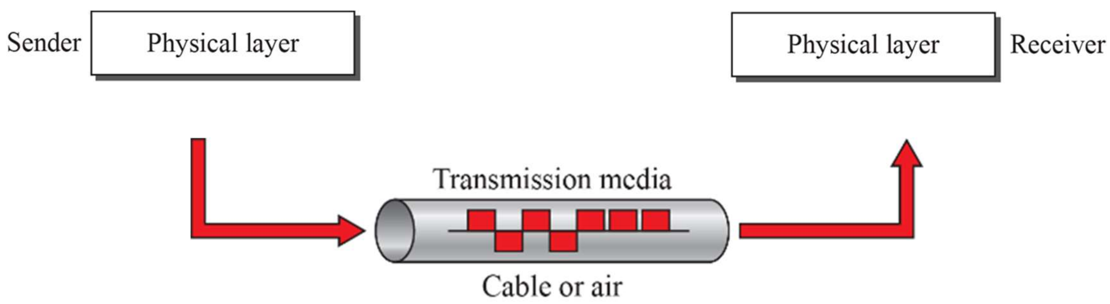
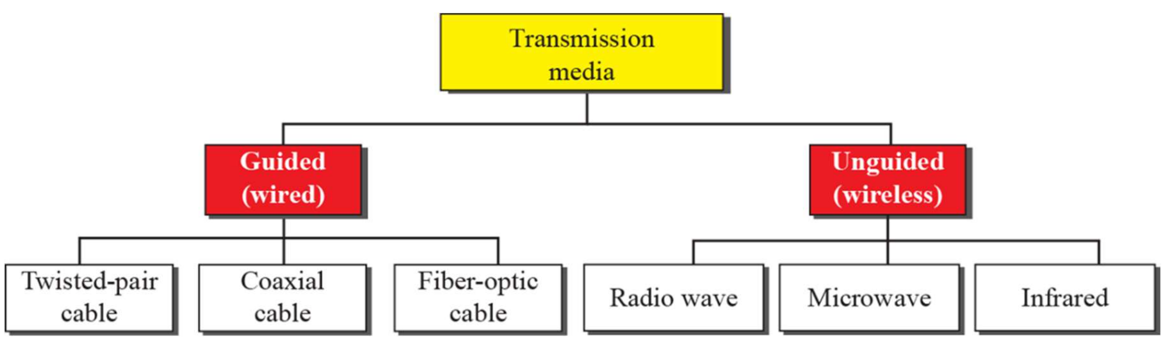


Fig7.1: Transmission media and physical layer

Types of Transmission Medium



GUIDED MEDIA

- **Guided media** provide a physical path for signals (e.g., twisted-pair, coaxial, fiber-optic cables).
- Signal is **confined within the medium**.
- **Twisted-pair cable**: two insulated copper wires twisted together.
- One wire carries the signal; the other serves as **ground reference**.
- **Receiver detects the difference** between the two wires.
- **Noise and crosstalk** can affect both wires and cause interference.

Twisted-Pair Cable

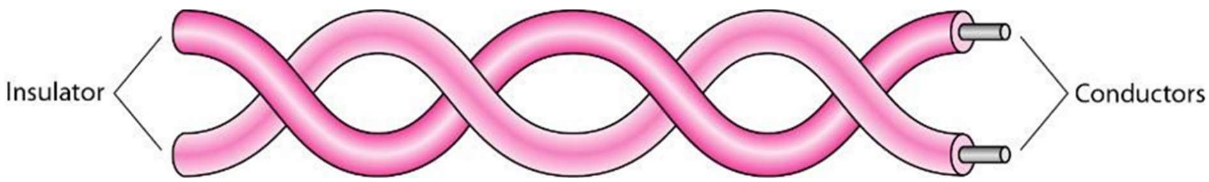


Fig7.3: Twisted-pair cable

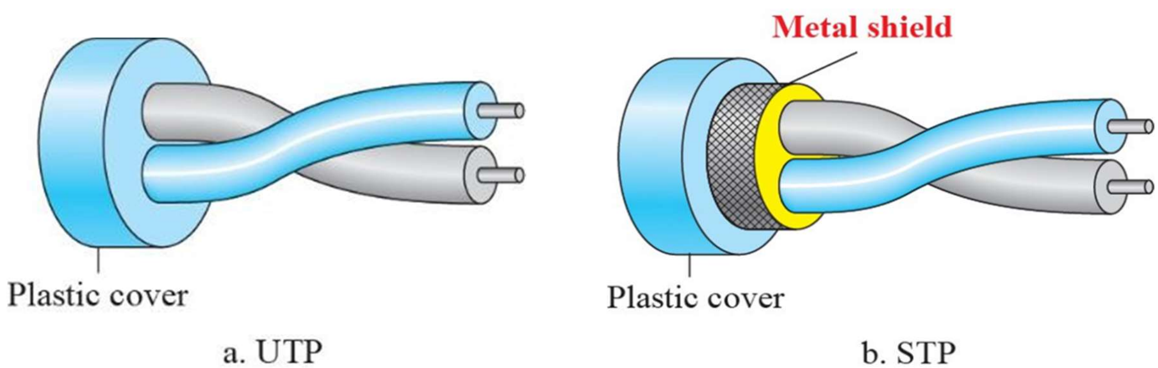


Fig7.4: Unshielded TP (UTP) and Shielded TP (STP) cables

Categories of unshielded twisted-pair cables

Category	*Specification*	*Data Rate (Mbps)*	*Use*
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs

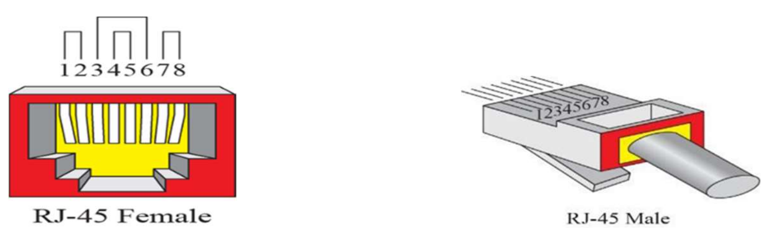


Fig7.5: UTP Connectors

Category	Specification	Data Rate (Mbps)	Use
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called *SSTP (shielded screen twisted-pair)*. Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

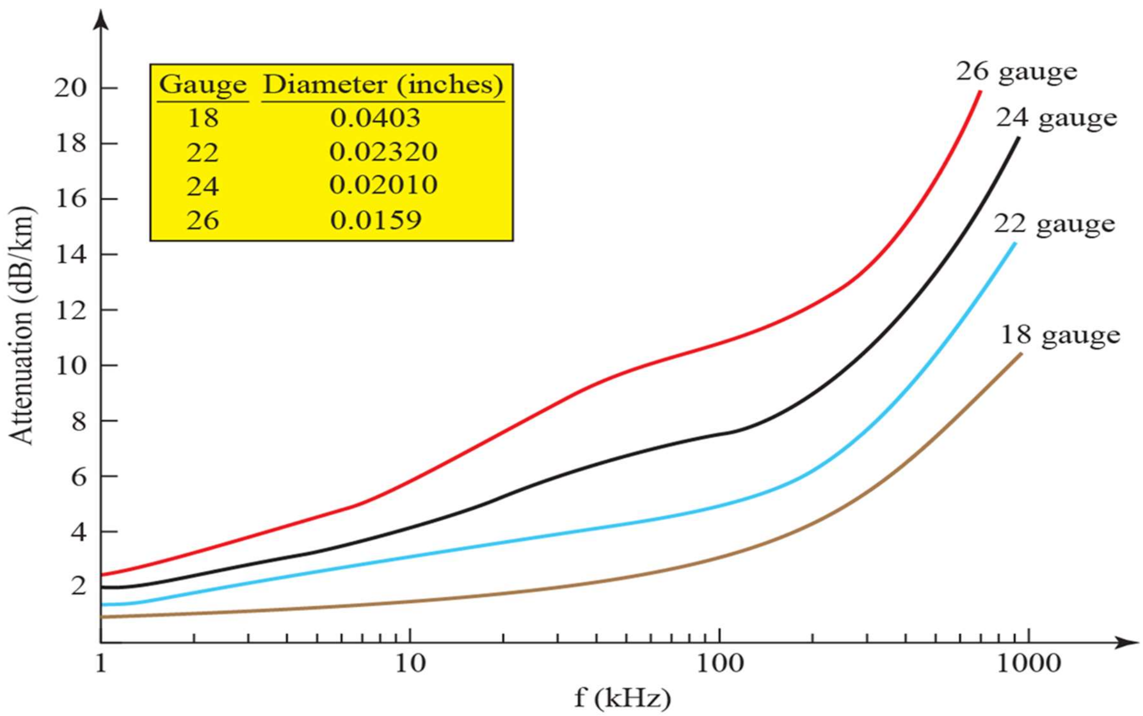


Fig7.6: UTP Performance

Coaxial Cable

- **Coaxial cable** carries **higher frequency signals** than twisted-pair.
- **Structure**: **central core conductor** (copper) → **insulating sheath** → **outer metal conductor** (foil/braid).
- **Outer conductor** acts as both **shield** against noise and **second conductor** to complete the circuit.

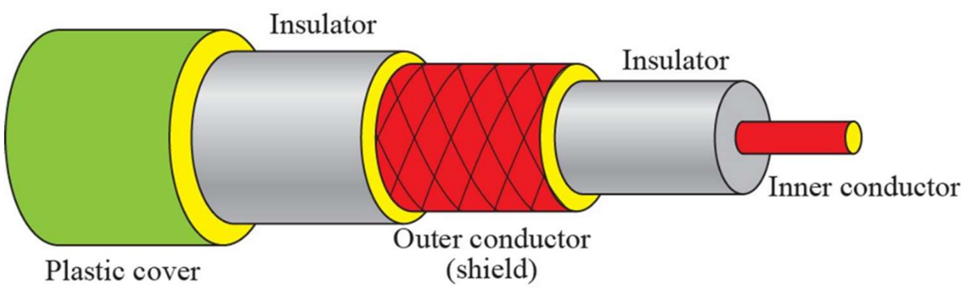


Fig7.7: Coaxial cable

Categories of coaxial cables

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

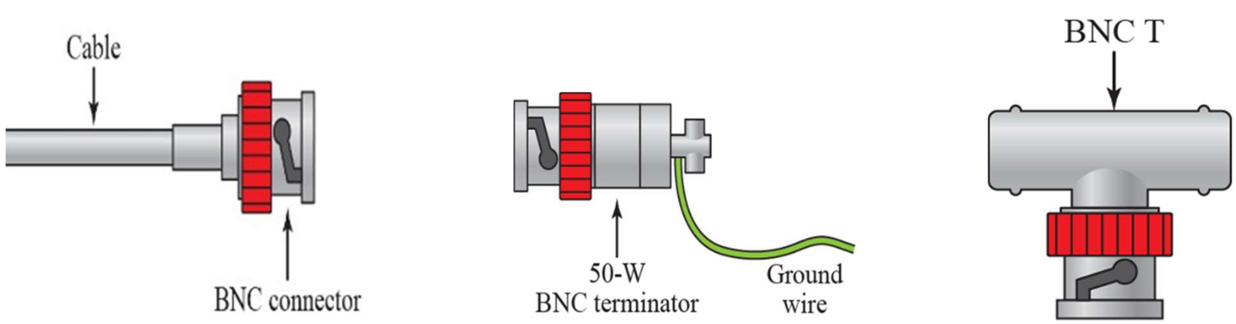


Fig7.8: Bayonet Neill-Concelman (BNC) connectors

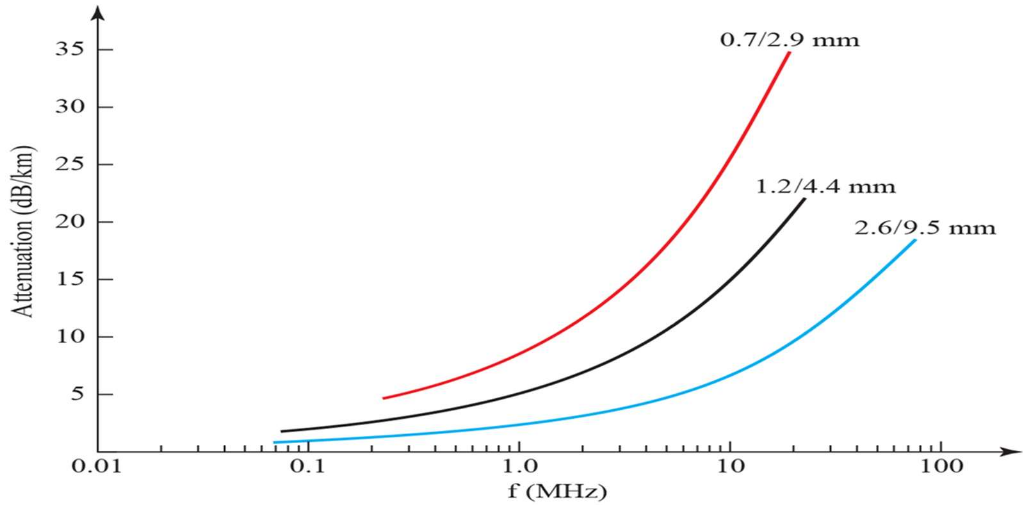
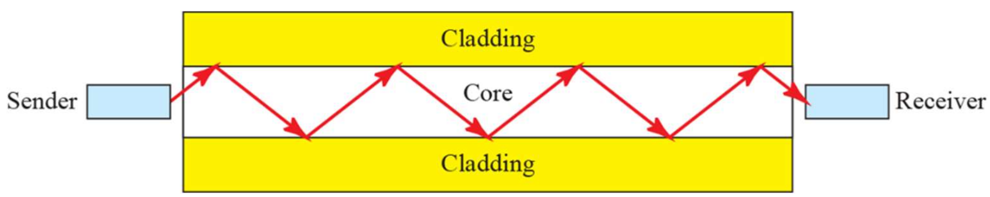


Fig7.9: Coaxial cable performance

Fiber-Optic Cable

- **Fiber-optic cable** is made of **glass or plastic**.
- It transmits signals using **light**.
- Understanding it requires knowledge of the **nature of light**.

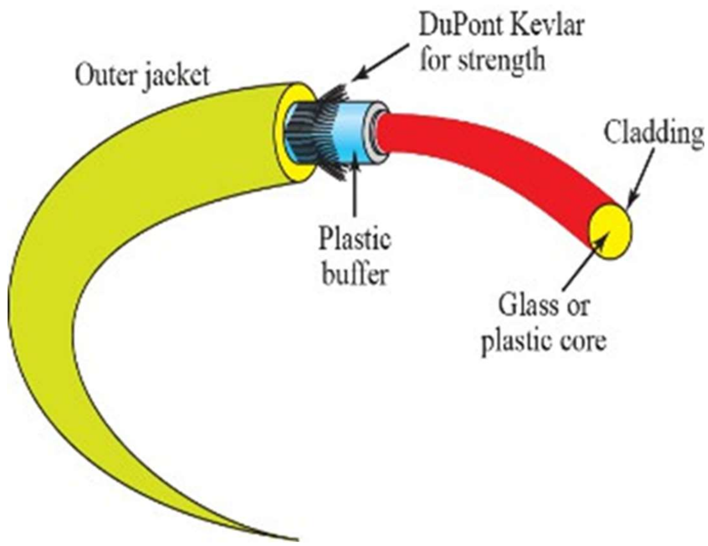


Figu7.11: Optical fiber

Types of Fiber-Optic Cable

Type	*Core (μm)*	*Cladding (μm)*	*Mode*
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

Table 7.3: Fiber types



Figu7.14: Fiber connection

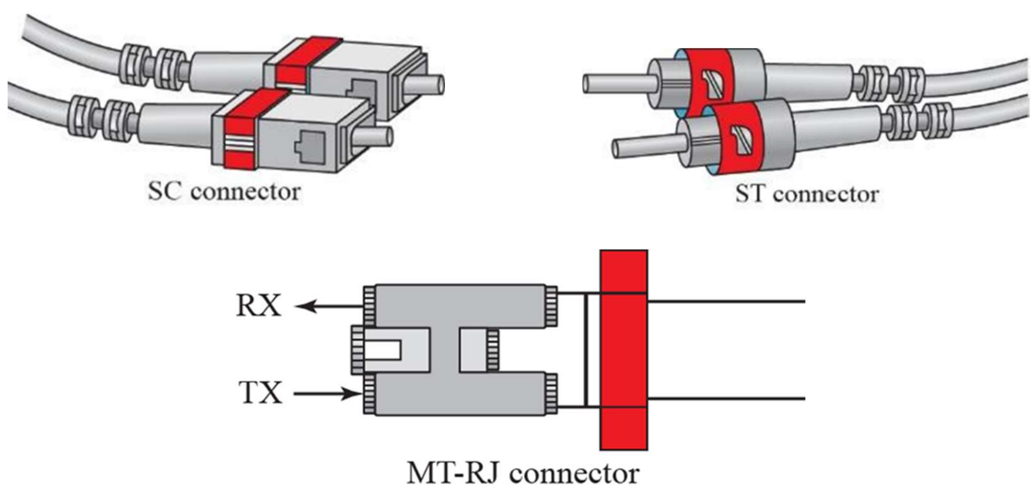


Fig7.15: Fiber-optic cable connector

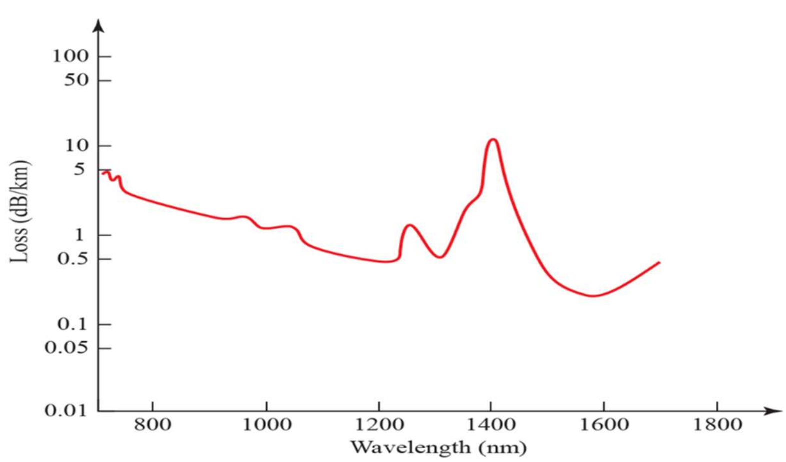


Fig7.16: Optical fiber performance

Optical Communication

- Fiber-optic system transmits **voice, video, or computer data**.
- **Analog signals** (like voice/video) are **converted to digital** using an **A/D converter**.
- **Computer data** is already digital.
- **Digital pulses** control a **light source**, turning it on and off rapidly for transmission.

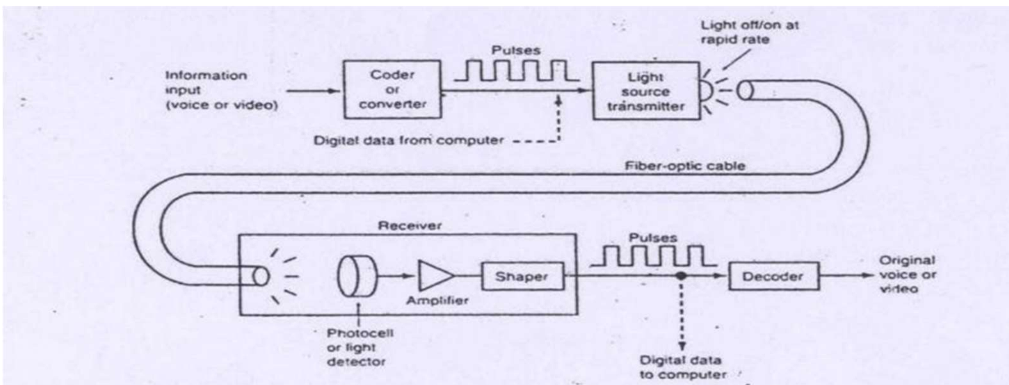


Fig. 13-1 Basic elements of a fiber-optic communications system.

Fiber Optic Communication System

- **Light Source**:
  - **LED or ILD**
  - Light output ∝ drive current
- **Source-to-Fiber Coupler**:
  - Works like a lens
  - Couples light into the optical fiber
- **Light Detector**:
  - **PIN diode** or **APD**
  - Converts light into current

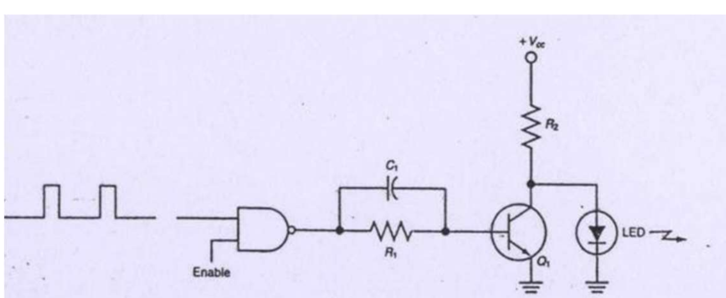
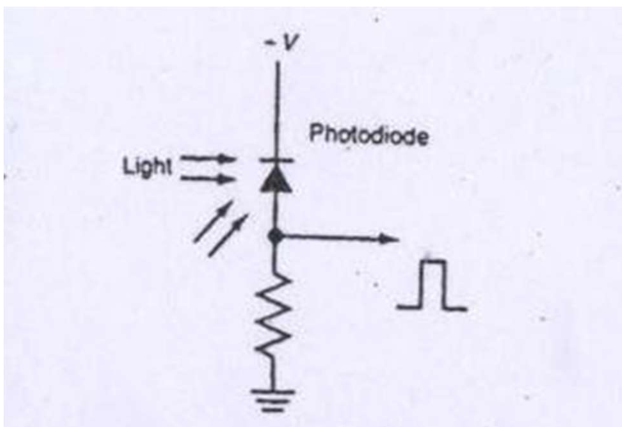


Fig13.23: Optical transmitter circuit using an LED

- **Digital Data Conversion**:
  - Converted to **serial pulse train** in **RZ or NRZ** format.
- **Light Transmitter**:
  - Made of **LED** and **driver circuit**.
- **Working**:
  - Binary pulses → **Logic gate** → **Transistor (Q1)** controls LED.
  - **Positive input** → NAND output = 0 → Q1 off → **LED ON**.
  - **Zero input** → NAND output = 1 → Q1 on → **LED OFF** (current bypassed).

- **Receiver is simple**.
- **Detector** senses light pulses → converts to **electrical signal**.
- Signal is **amplified** and **reshaped** to original digital data.
- Main component: **Photodiode** (light sensor).
- Photodiode = **silicon PN junction**, **reverse-biased**.



13-26 How a photodiode converts light into voltage pulses.

UNGUIDED MEDIA

- **Unguided Medium**: No physical conductor used.
- Known as **wireless communication**.
- **Signals travel through free space**.
- Can be received by **any compatible device**.

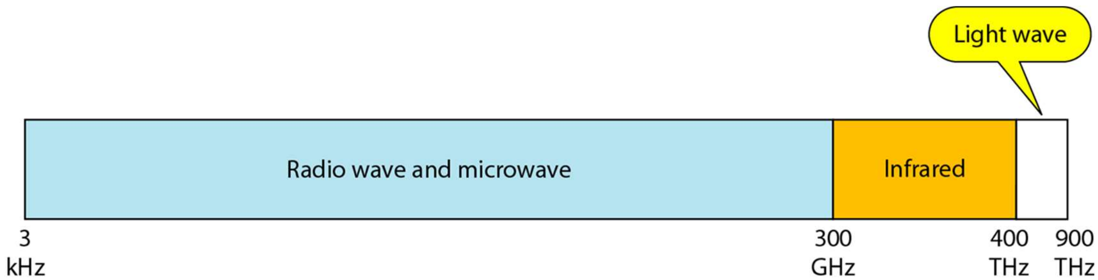


Fig7.17: Electromagnetic spectrum for wireless communication

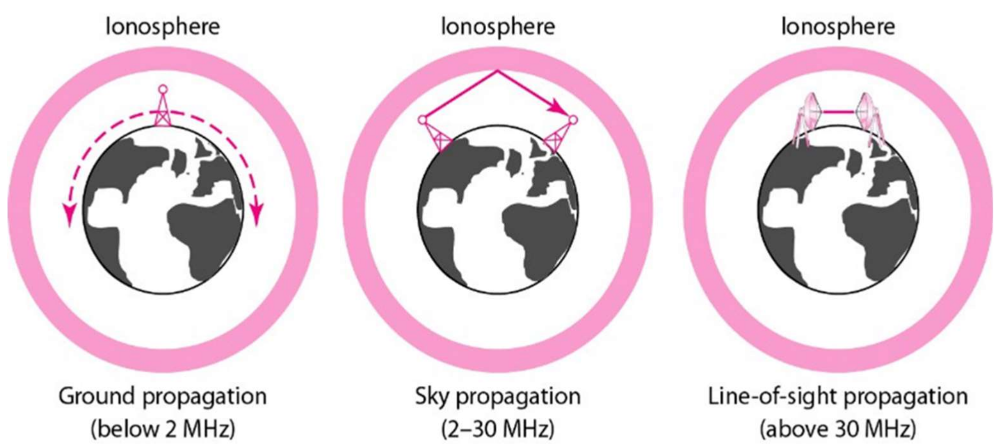


Fig7.18: Propagation methods

Band	*Range*	*Propagation*	*Application*
very low frequency (VLF)	3–30 kHz	Ground	Long-range radio navigation
low frequency (LF)	30–300 kHz	Ground	Radio beacons and navigational locators
middle frequency (MF)	300 kHz–3 MHz	Sky	AM radio
high frequency (HF)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft
very high frequency (VHF)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
ultrahigh frequency (UHF)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
superhigh frequency (SF)	3–30 GHz	Line-of-sight	Satellite
extremely high frequency (EHF)	30–300 GHz	Line-of-sight	Radar, satellite

Table 7.4: Bands

Radio Waves

- No sharp boundary between **radio waves** and **microwaves**.
- **Radio waves:** 3 kHz to 1 GHz frequency range.
- **Microwaves:** 1 GHz to 300 GHz frequency range.
- Better to classify by **wave behavior** than just frequency.

Fig7.19: Omnidirectional antenna

Microwaves

- **Microwaves**: Electromagnetic waves with **1 to 300 GHz** frequency.
- They are **unidirectional** (travel in a narrow beam).
- Transmitting antennas can **focus microwaves narrowly**.
- Sending and receiving antennas must be **aligned**.
- Advantage: multiple pairs of antennas can operate **without interfering** with each other.

Fig7.20: Unidirectional antenna

Infrared

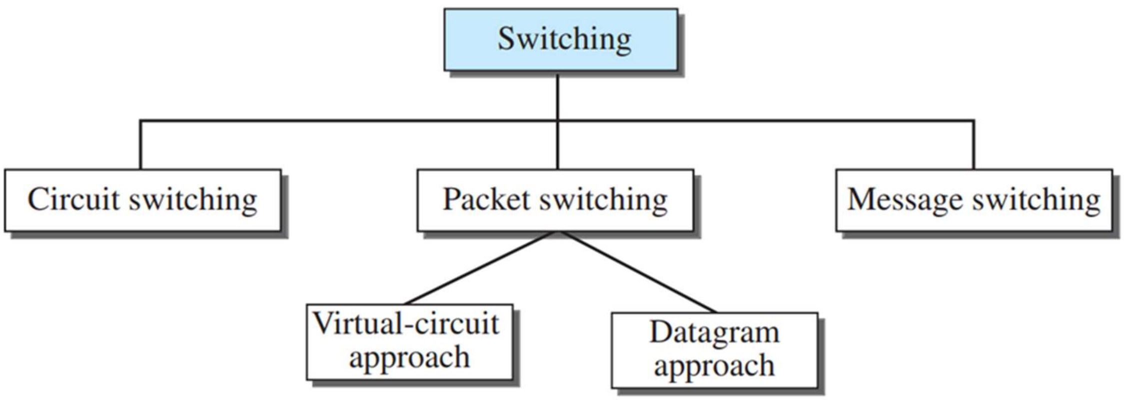
- **Infrared waves**: Frequencies from **300 GHz to 400 THz** (wavelengths 1 mm to 770 nm).
- Used for **short-range communication**.
- **Cannot penetrate walls,** so signals stay within a room.
- This prevents **interference** between nearby systems (e.g., infrared remotes).
- Example: Your remote control doesn't interfere with your neighbor's remote.

NETWORK SWITCHING

Switching Mechanism for Data Transfer

➔ **Switching Mechanism for Data Transfer** refers to the method used to route data from source to destination in a network. It determines how data travels through the network. The main types are:

1. **Circuit Switching** – Establishes a dedicated path before communication starts.
2. **Packet Switching** – Data is broken into packets and sent independently.
3. **Message Switching** – Entire messages are stored and forwarded hop-by-hop.



Switching and TCP/IP Layers

1. **Physical Layer**
    - **Type of Switching**: **Circuit Switching** only.
    - **Explanation**: This layer deals with the actual transmission of electrical or optical signals. There are no packets here—only raw data signals. Switching happens by connecting wires or circuits, like in traditional telephone lines.
2. **Data-Link Layer**
    - **Type of Switching**: **Packet Switching** (but here, packets are called **frames** or **cells**).
    - **Approach**: **Virtual Circuit**.
    - **Explanation**: This layer handles communication between devices on the same network. The switch stores and forwards frames based on predefined logical paths.

3. **Network Layer**
    - **Type of Switching**: **Packet Switching**.
    - **Approaches**:
        - **Datagram** (current Internet model): Packets are sent independently and may take different routes.
        - **Virtual Circuit** (being considered for future use): A logical path is set up, and all packets follow it.
    - **Explanation**: This layer handles routing between different networks. It decides the best path for data to travel from source to destination.
4. **Application Layer**
    - **Type of Switching**: **Message Switching** (conceptual).
    - **Explanation**: Communication happens by sending complete messages, like sending an email. There's no real-time connection—each message is stored and then forwarded. This is mostly theoretical now and not used in modern networks.

Circuit Switching

- **Dedicated path** between two stations
- Requires **switching and channel capacity**
- Needs **routing intelligence**
- **Inefficient**: unused capacity is wasted
- **Setup takes time**
- Designed for **voice traffic**
- **Examples**: Telephone networks, ISDN

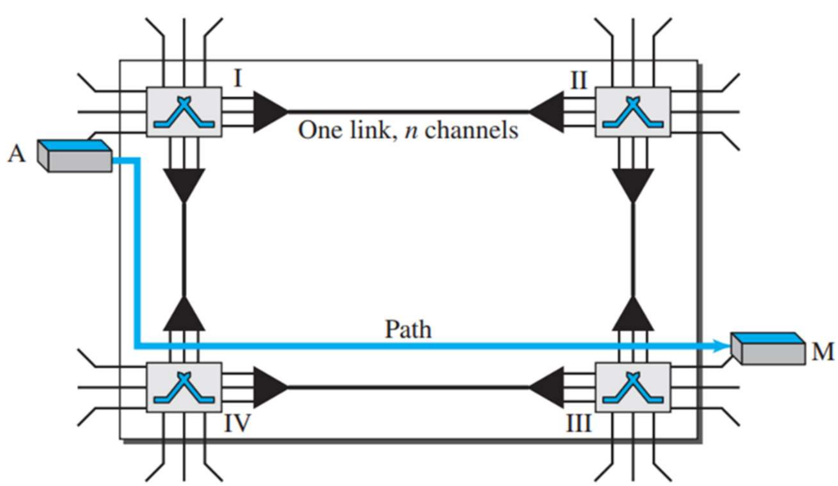


Fig: A trivial circuit-switched network

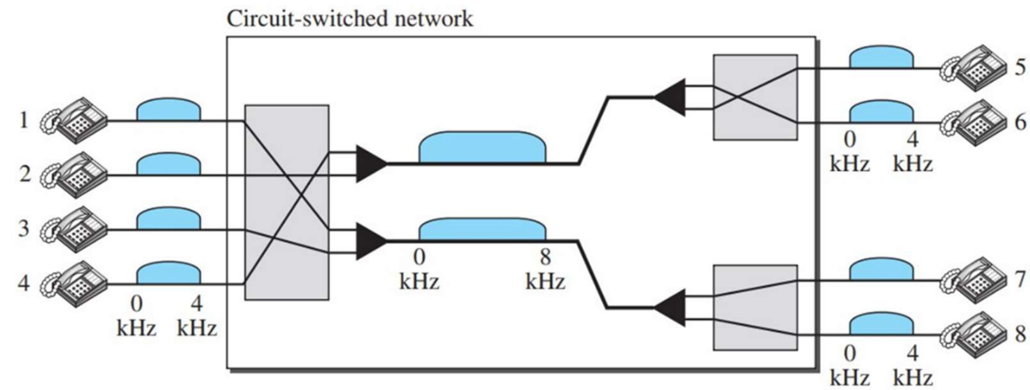


Fig: Circuit-switched network used in Example 8.1

Example 8.1

As a trivial example, let us use a circuit-switched network to connect eight telephones in a small area. Communication is through 4-kHz voice channels. We assume that each link uses FDM to connect a maximum of two voice channels. The bandwidth of each link is then 8 kHz. Figure 8.4 shows the situation. Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6. Of course the situation may change when new connections are made. The switch controls the connections.

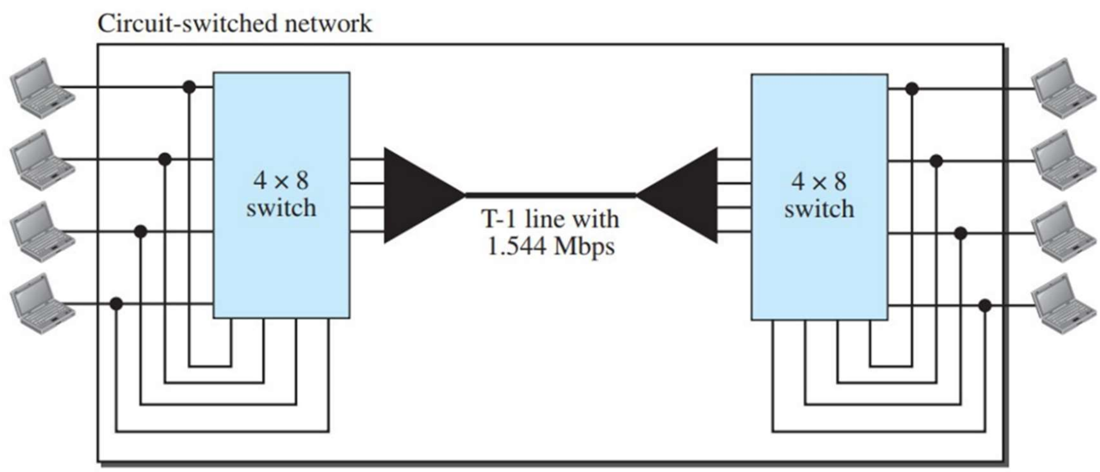
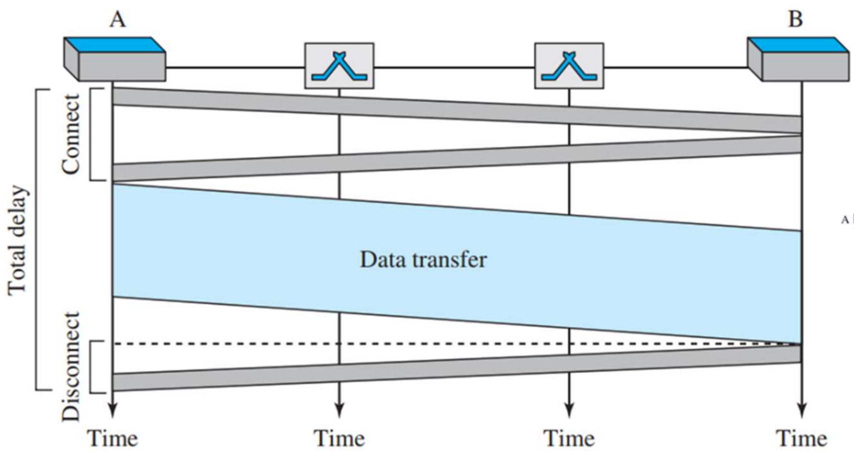


Fig: Circuit-switched network used in Example 8.2

Example 8.2

As another example, consider a circuit-switched network that connects computers in two remote offices of a private company. The offices are connected using a T-1 line leased from a communication service provider. There are two 4 × 8 (4 inputs and 8 outputs) switches in this network. For each switch, four output ports are folded into the input ports to allow communication between computers in the same office. Four other output ports allow communication between the two



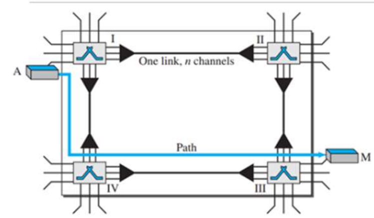
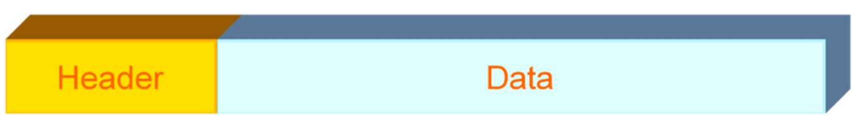


Fig: Delay in a circuit-switched network

Packet Switching

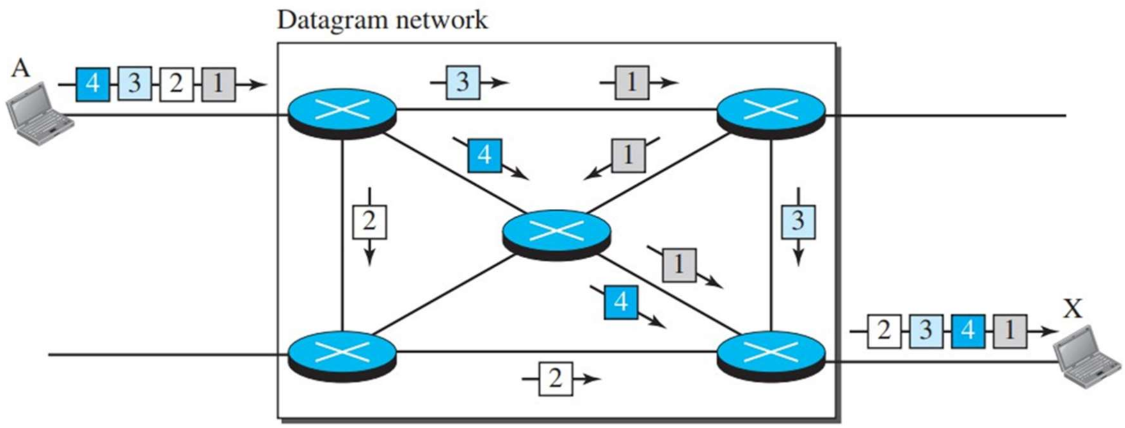
- **Messages are split into packets** (small fixed-size segments)
- **Packets** allow **faster routing**
- **Packet structure** includes a **header** with control info (source, destination, IDs, etc.)
- Packets move **node to node** through routing
- Uses **store-and-forward** technique at each node
- **Minimal storage** needed at switches/nodes



Packet Switching Advantages

- **Packetization** allows short messages to pass without waiting for long ones
- **Efficient line usage**: multiple packets share the same link over time
- **Packets are queued and sent quickly**
- **Works even in busy networks**, though delivery may slow
- **Priority levels** can be assigned to packets

Datagram packet Switching



- **Each packet is switched independently**
- **Header includes destination address**
- **No pre-allocated resources**
- **Routes can change during transmission**

Datagram packet switching

- **Switch uses routing table** based on **destination address**
- **Destination address stays the same** throughout the packet's journey

Destination address	Output port
1232	1
4150	2
⋮	⋮
9130	3

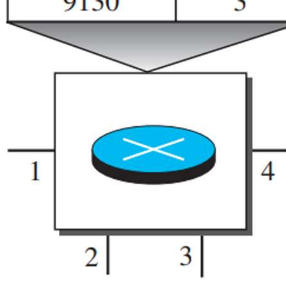


Fig: Routing table in a datagram network

Delay in Datagram packet switching

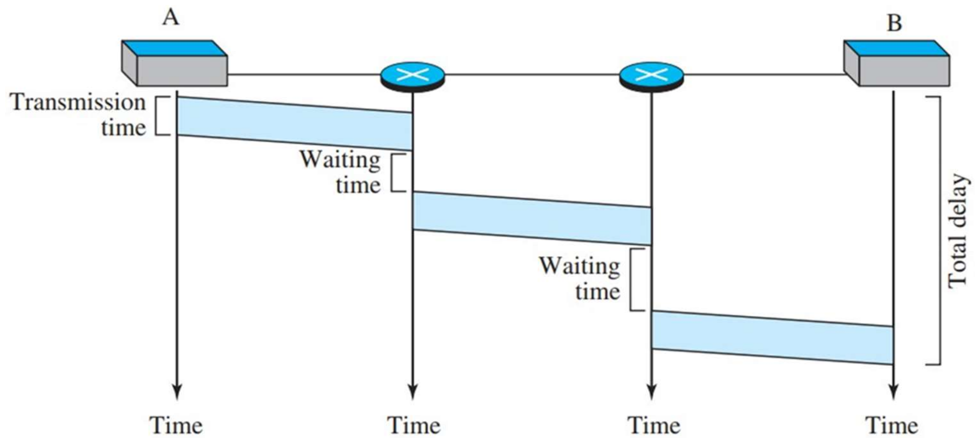


Fig: Delay in a datagram network

- **Total delay** = **3T** (transmission) + **3τ** (propagation) + **w_1 + w_2** (waiting times)
- **Processing time is ignored**

$$\textbf{Total delay} = 3T + 3\tau + w_1 + w_2$$

Virtual-Circuit Packet Switching

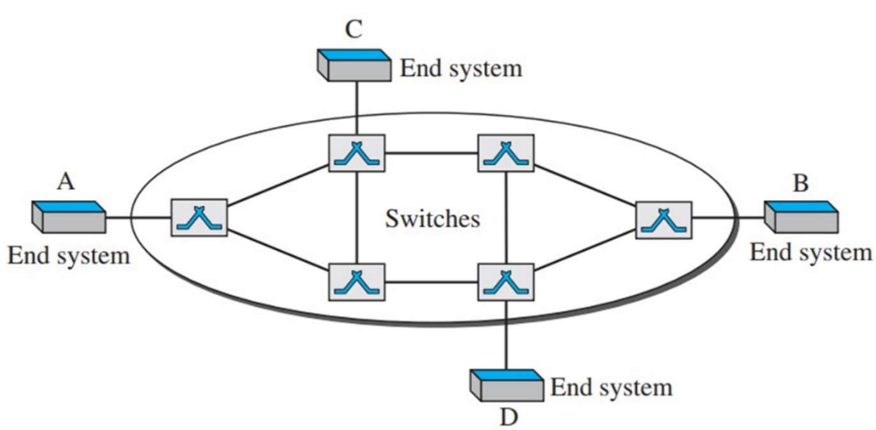


Fig: Virtual-circuit network

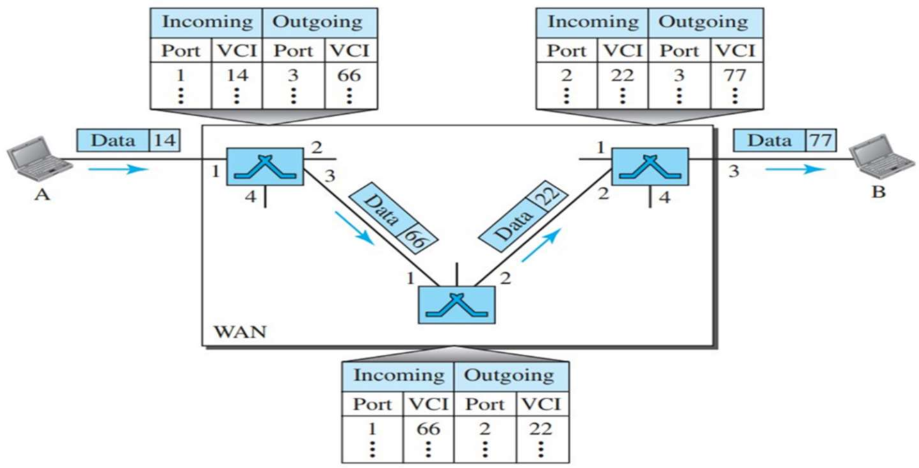
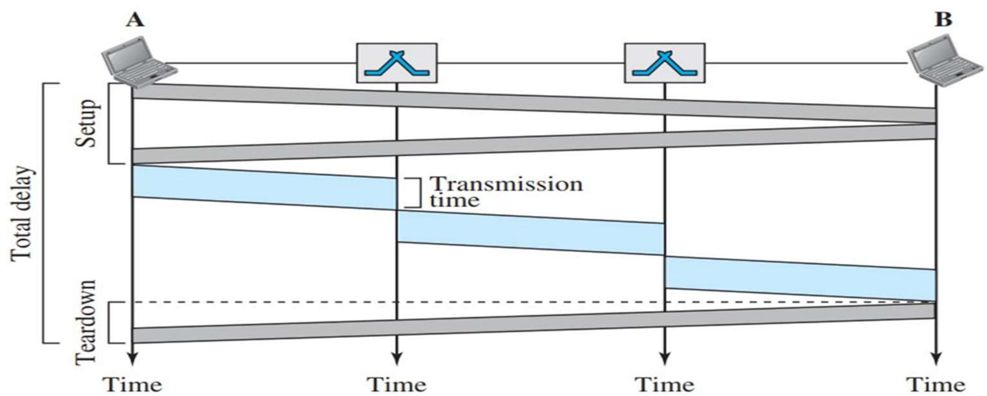


Fig: Source-to-destination data transfer in a virtual-circuit network

Delay in Virtual-Circuit Packet Switching



- Packet travels through two switches (routers)
- Delays involved:
  - **3 Transmission times** = 3T
  - **3 Propagation times** = 3τ
  - **Setup delay**: includes transmission and propagation in **two directions**
  - **Teardown delay**: includes transmission and propagation in **one direction**
- Processing time at switches is ignored
- **Total delay time** includes **transmission, propagation, setup, and teardown delays**

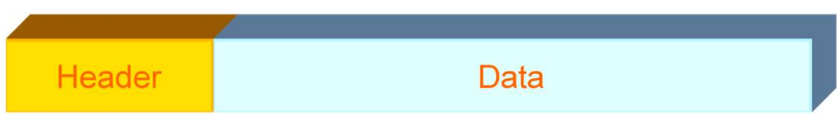
Total delay = 3T + 3τ + Setup delay + Teardown delay

Datagram vs Virtual-Circuits Switching

Datagram	Virtual Circuits
Connectionless service	Connection-oriented service
More flexible routing	Faster forwarding (no routing decisions)
Less reliable	Highly reliable
High efficiency, but more delay	Lower efficiency, less delay
Easy and cost-effective to implement	Costly to implement
More overhead per packet (includes full address)	Less overhead (setup done once)

Message Switching

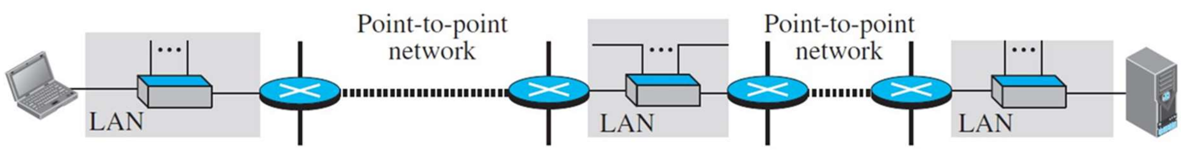
- **No dedicated path** required between sender and receiver
- **No real-time interaction** needed; destination status not checked beforehand
- **Each message is independent** and includes full destination address
- **Messages are stored** at each node before being forwarded (store-and-forward)
- **Accepts all traffic**, but **delivery is slower** than circuit switching
- **Circuit switching may reject traffic**, but message switching does not



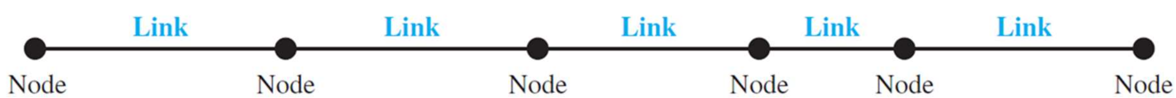
DATA-LINK LAYER

Nodes and Links

- A data unit travels through many networks (LANs/WANs) on the Internet.
- Routers connect these networks.
- Routers take data from one network and send it to another.
- **Nodes** = End devices + Routers
- **Links** = Networks between nodes



a. A small part of the Internet



b. Nodes and links

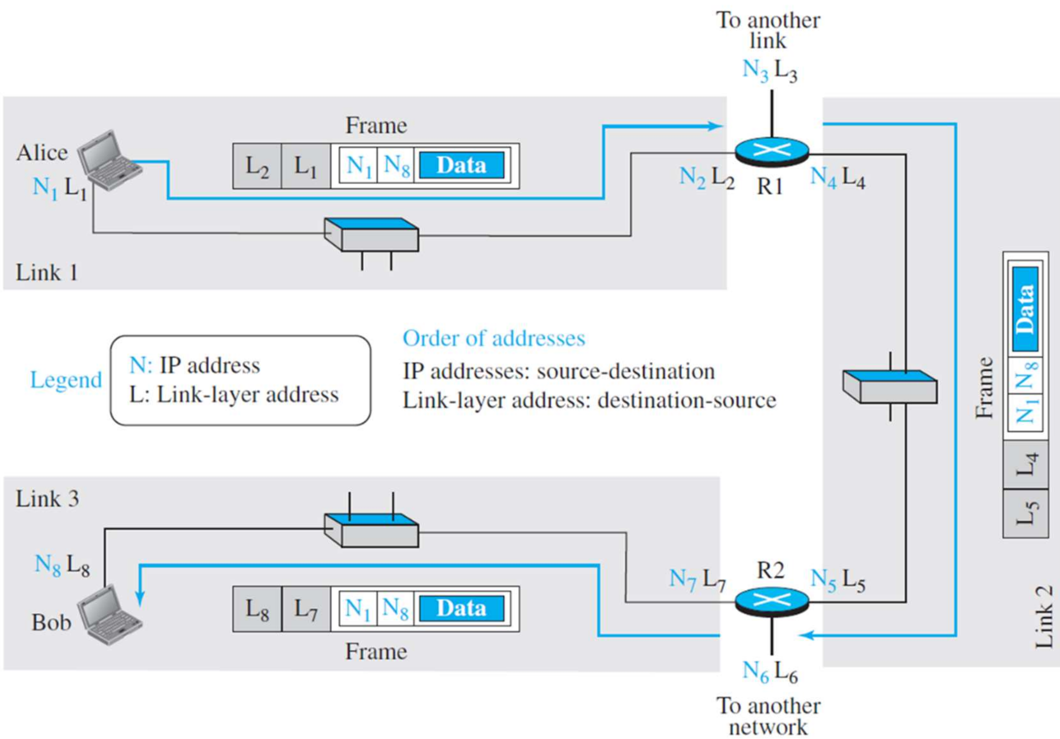
Communication in the Data-Link Layer

- **Data-link layer** handles **node-to-node** communication.
- It delivers a packet from one node to the next in the path.
- **Source**: encapsulates the packet.
- **Destination**: decapsulates the packet.
- **Routers (intermediate nodes)**: both encapsulate and decapsulate.
- A packet at the **data-link layer** is called a **frame**.

Addressing Mechanism for Data-Link Layer

- Data units can take **different paths** from source to destination.
- **IP addresses** (network layer) define source and destination — they **don't change** and **don't control path**.
- To move data **within a network**, **MAC address** (data-link layer) is used.
- MAC address = **Physical address**.

Figure 9.5 *IP addresses and link-layer addresses in a small internet*



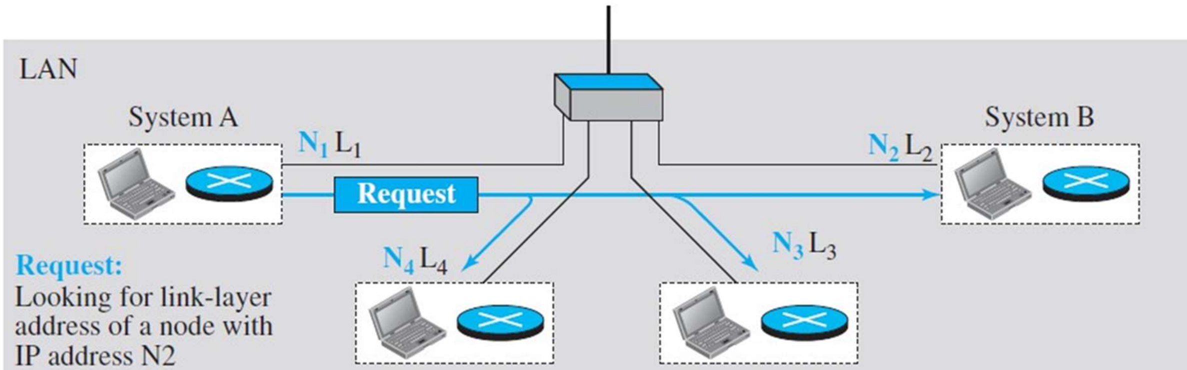
Link Layer Addresses

- **Unicast**: One-to-one communication
  - Example: **`A3:34:45:11:92:F1`**
  - Unique address for each device/interface
- **Multicast**: One-to-many (within local network)
  - Similar format as unicast
  - Second hex digit must be **even**
  - Example: **`A2:34:45:11:92:F1`**
- **Broadcast**: One-to-all (within the link)
  - All bits are 1s
  - Example: **`FF:FF:FF:FF:FF:FF`**

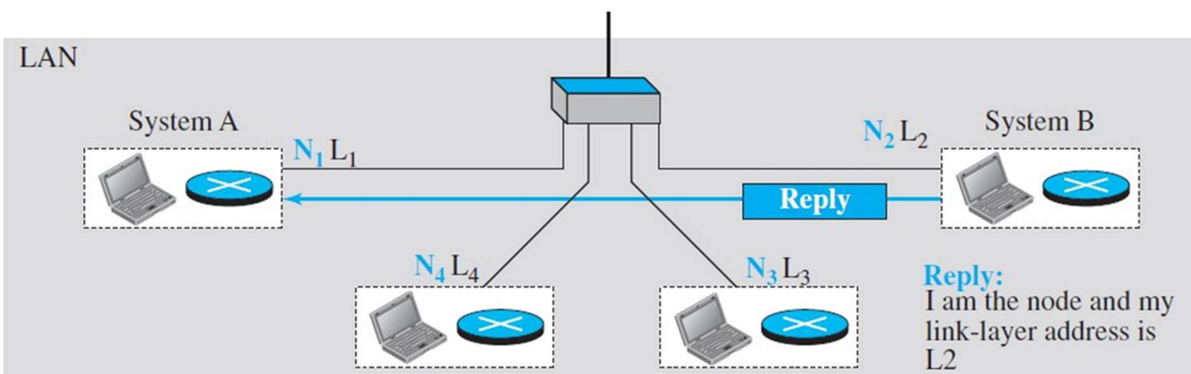
Address Resolution Protocol (ARP)

- **Switch**: Data-link layer device; uses **MAC addresses** to forward data.
- **Router**: Network layer device.
- **ARP (Address Resolution Protocol)** maps **IP address → MAC address**.
- If MAC is unknown, ARP sends a **broadcast** to all nodes.
- Only the target device replies with its **MAC address**.

Figure 9.7 *ARP operation*



a. ARP request is broadcast

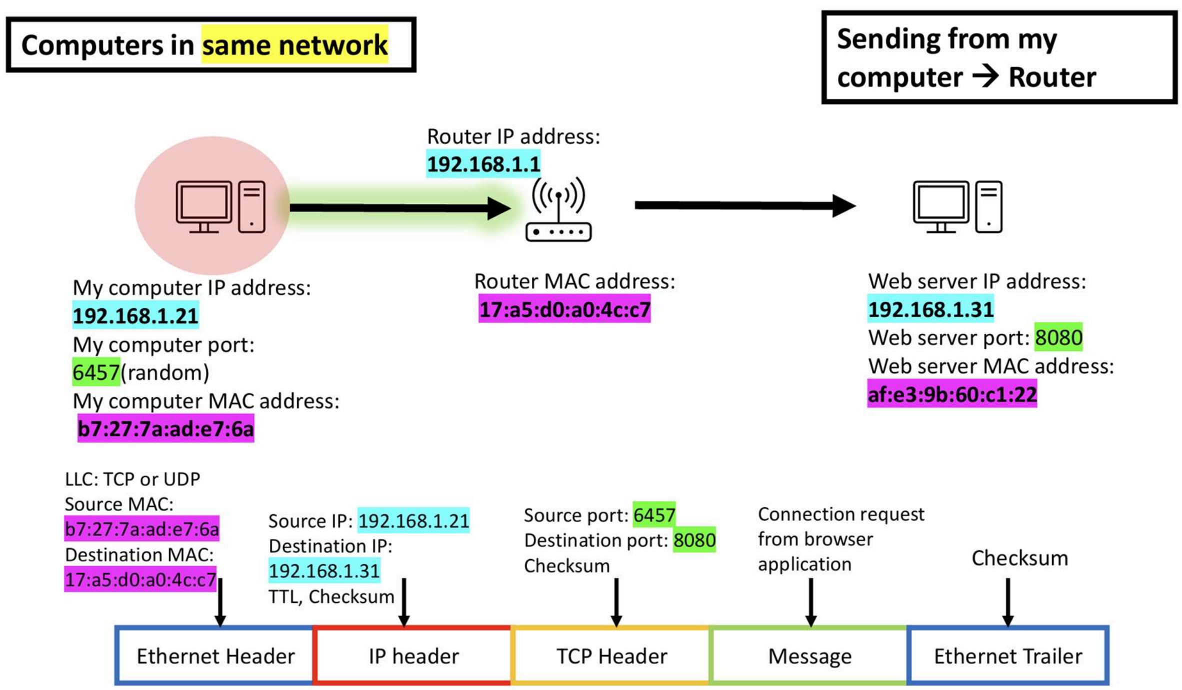


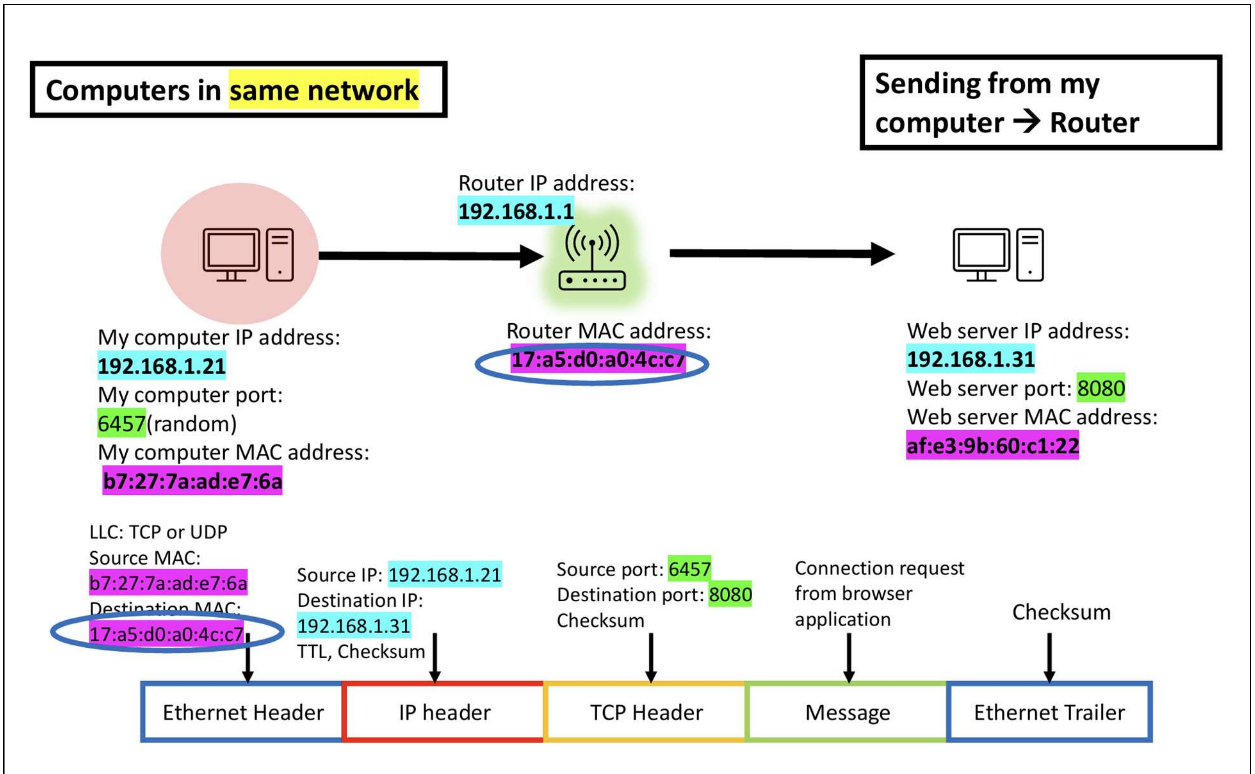
b. ARP reply is unicast

Example:

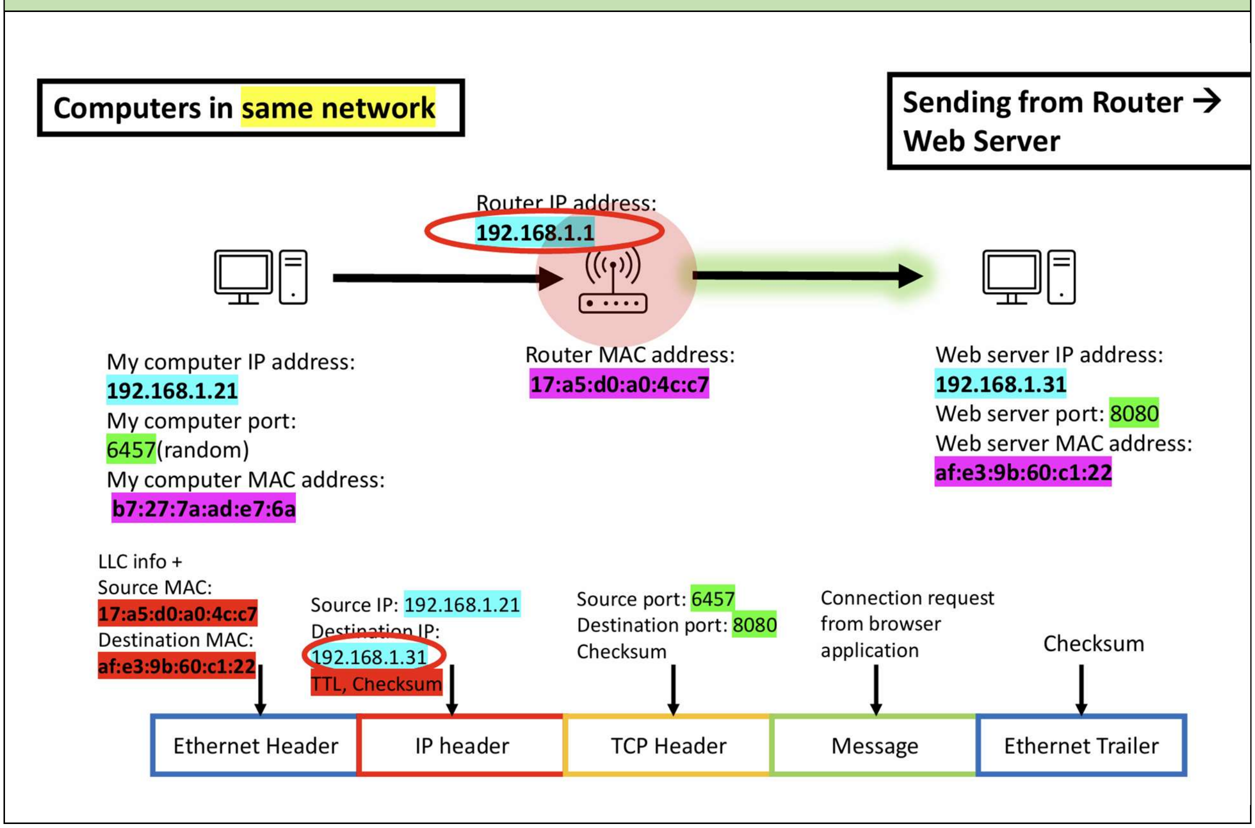
- PC communicating with a web server in the same network (all TCP/IP layers involved).
- **MAC addresses** change at each link.
- **IP addresses** stay the same from source to destination.
- **Ports** (transport layer) help identify the correct service (e.g., web server, email).
- **Encapsulation/decapsulation** happens at every node on the path.

Sending from my computer to Router:

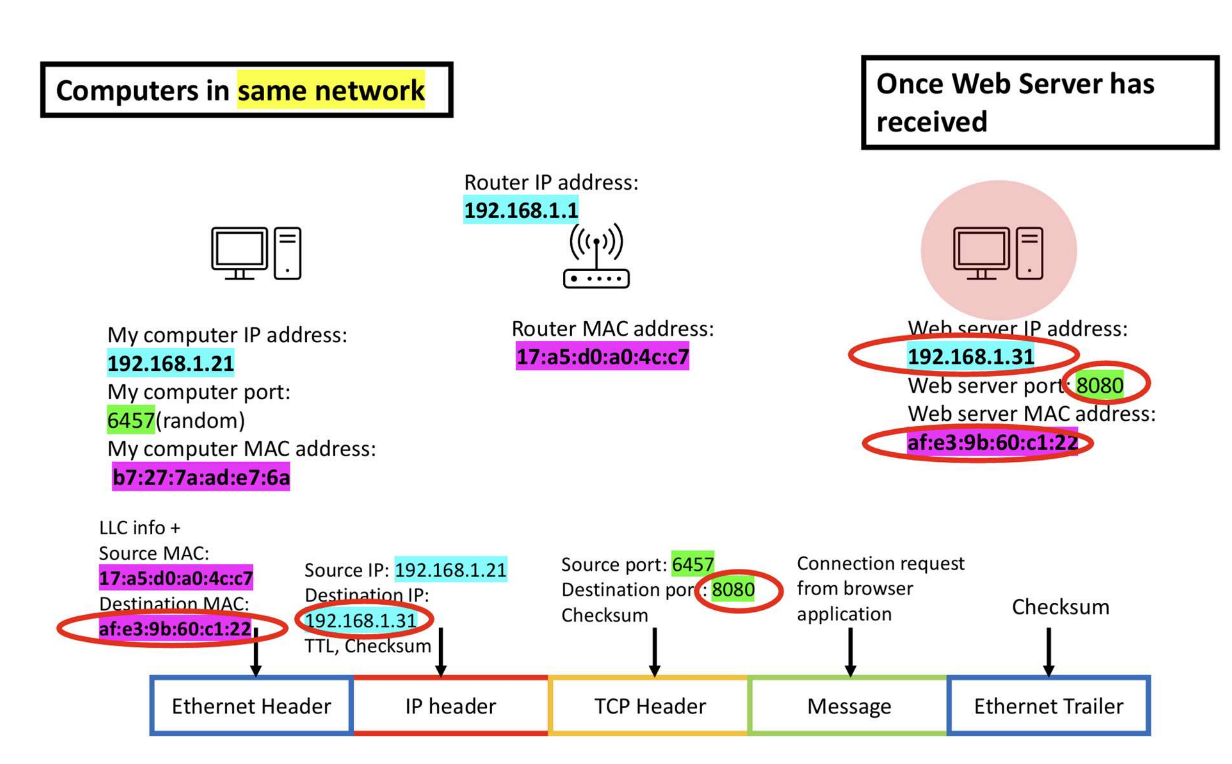




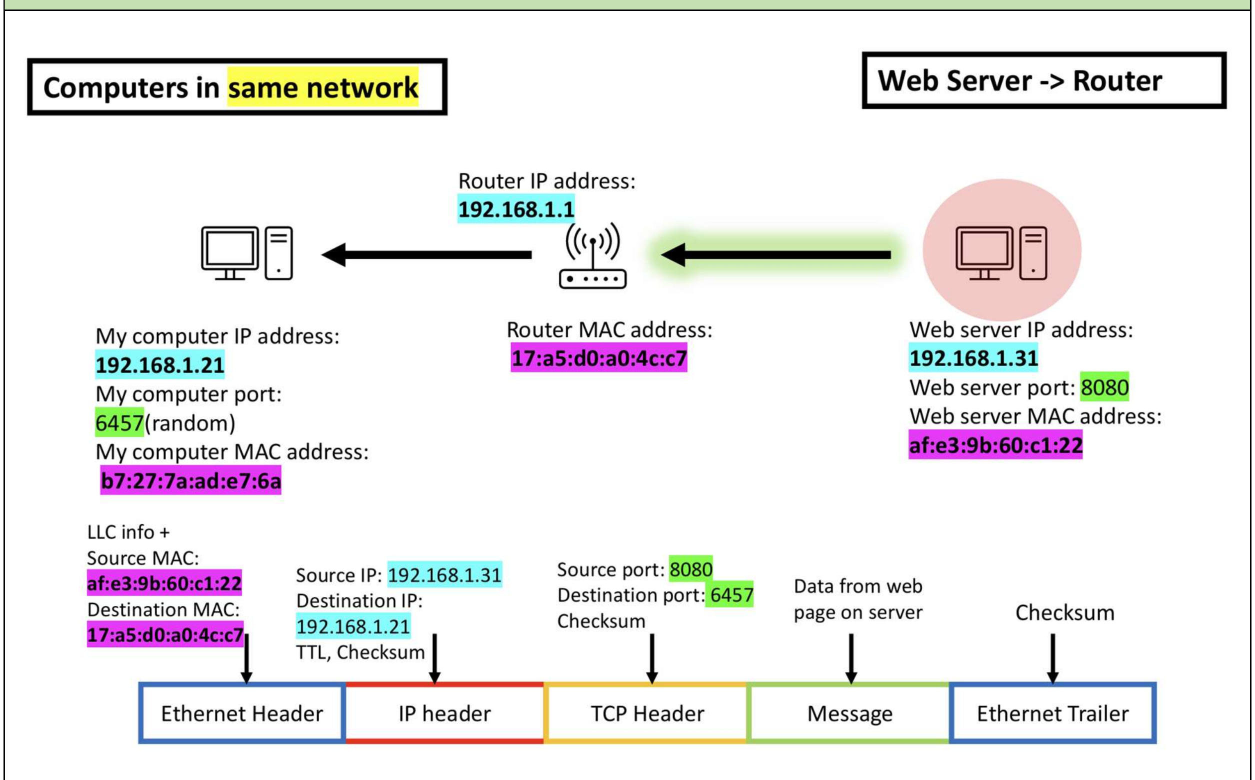
Sending from Router to Web Server:



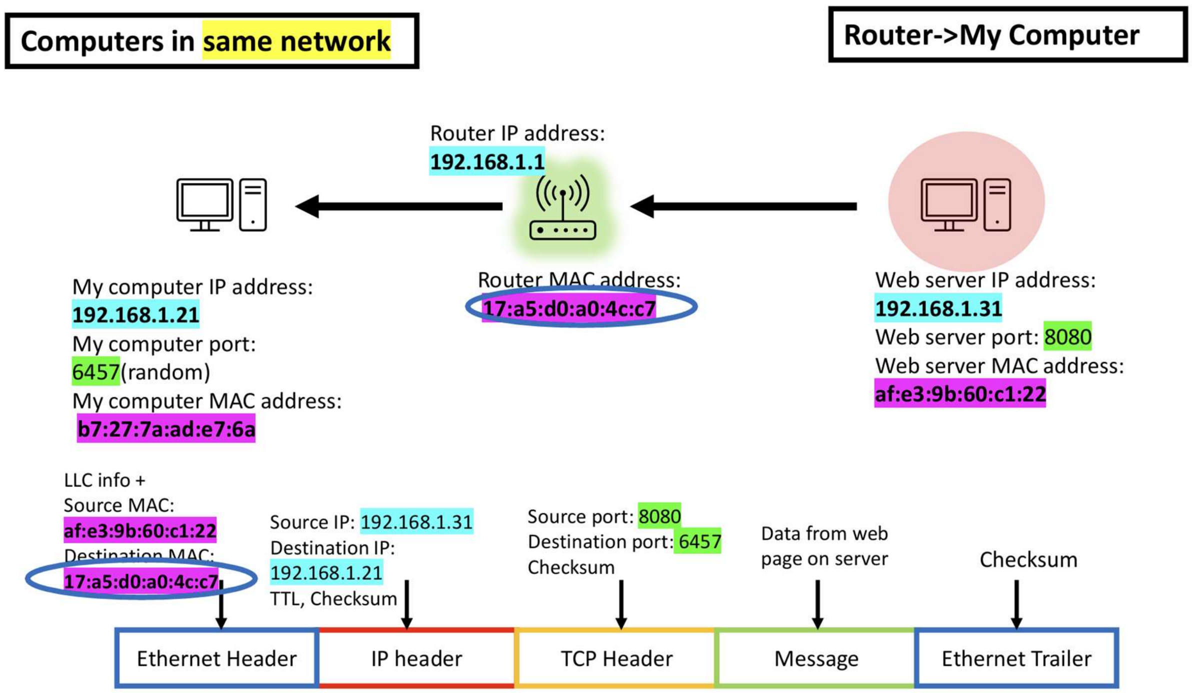
Once Web Server has recived:

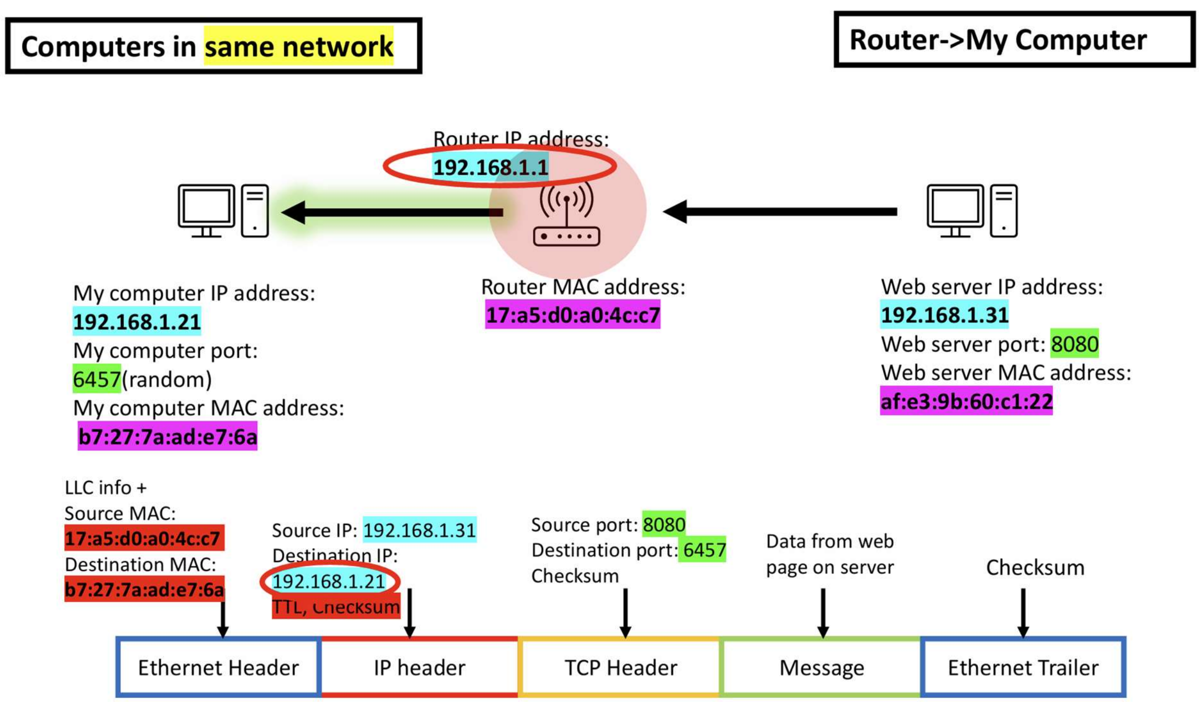


Sending from Web Server to Router:



Sending from Router to My Computer:





Once My Computer has recived: